DERMATOSURGRY

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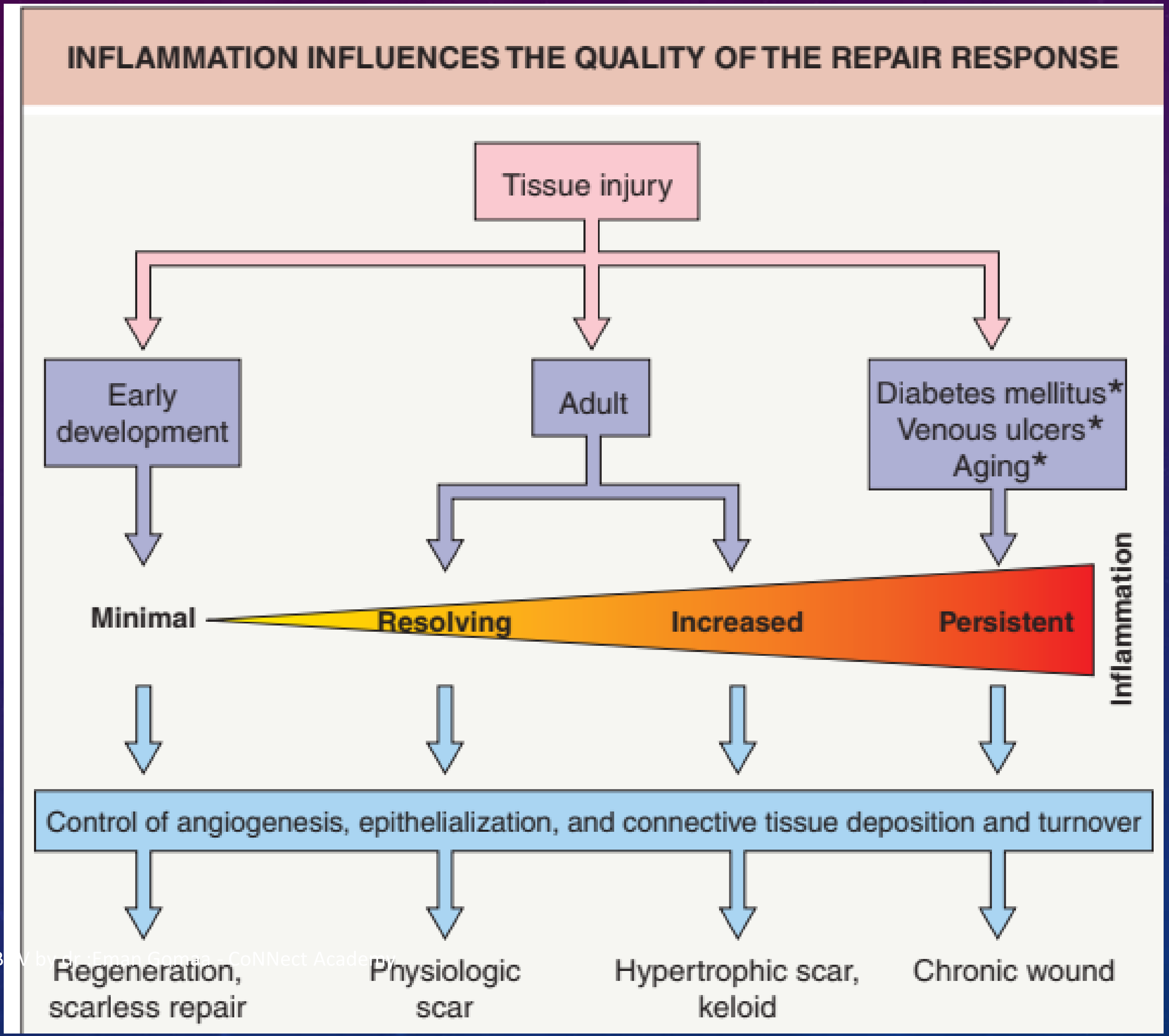
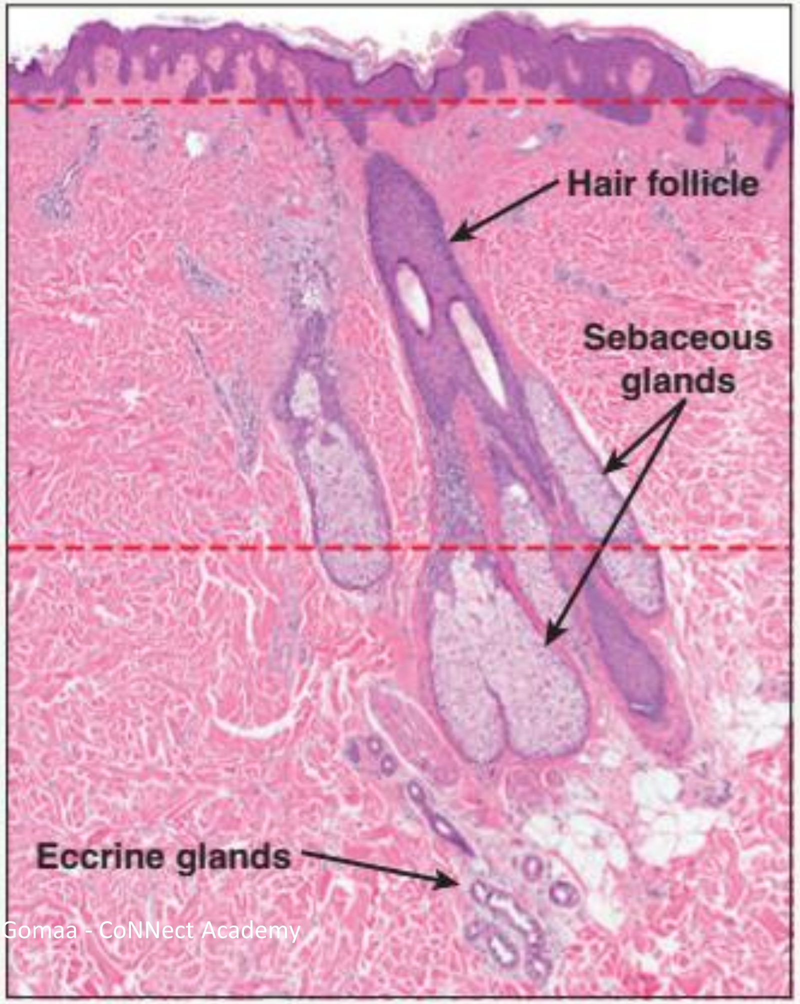
WOUND HEALING


INFLAMMATION INFLUENCES THE QUALITY OF THE REPAIR RESPONSE
Tissue injury
Early development
Adult
Diabetes mellitus*
Venous ulcers*
Aging*
Minimal
Resolving
Increased
Persistent
Inflammation
Control of angiogenesis, epithelialization, and connective tissue deposition and turnover
Regeneration, scarless repair
Physiologic scar
Hypertrophic scar, keloid
Chronic wound

WOUND DEPTH AND IMPLICATIONS FOR THE QUALITY OF SKIN REPAIR



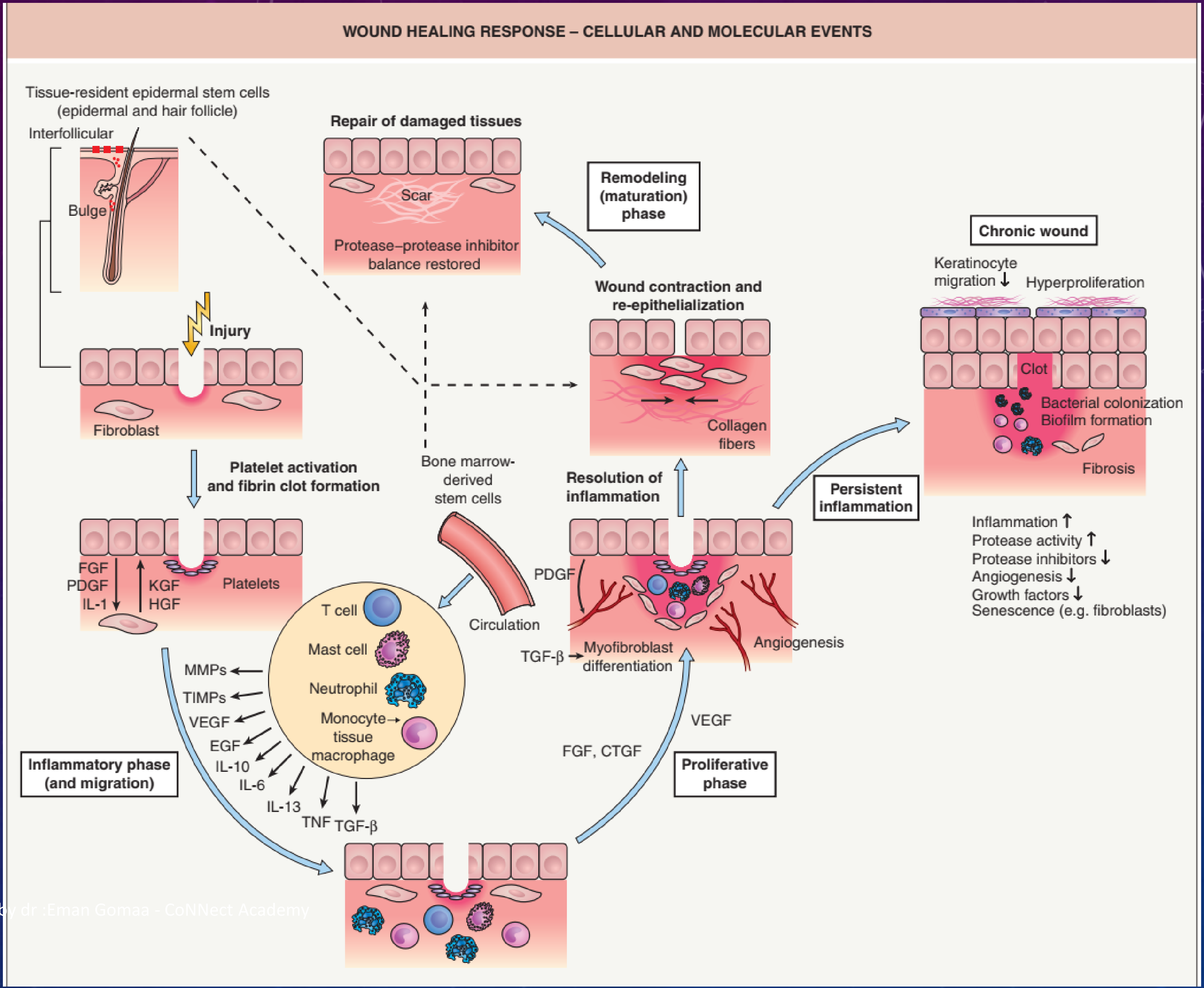
Erosion
- epidermal regeration
- no scar formation

Partial-thickness wound
- re-epithelialization occurs from the wound edges and remaining adnexal structures
- scar formation

Full-thickness wound
- re-epithelialization occurs only from the wound edges
- scar formation
- wound contraction

CHEMICAL MEDIATORS OF INFLAMMATION THAT PLAY A ROLE IN WOUND HEALING	
Chemical mediator	**Action**
Histamine	Increased vascular permeability
Serotonin	Stimulation of fibroblast proliferation Cross-linking of collagen molecules
Kinins	Increased vascular permeability
Prostaglandins	Increased vascular permeability
	Sensitize pain receptors
	Increased synthesis of GAGs
Complement	Increased vascular permeability
	Increased phagocytosis
	Enhanced bacterial lysis
	Mast cell and basophil activation

GROWTH FACTORS INVOLVED IN WOUND HEALING			
Growth factor	**Abbreviation**	**Source**	**Functions**
Platelet-derived growth factor	PDGF	Platelets, keratinocytes, fibroblasts, endothelial cells, perivascular cells	Fibroblast proliferation, chemotaxis & collagen metabolism; angiogenesis
Transforming growth factor-β	TGF-β	Platelets, keratinocytes, fibroblasts, endothelial cells, macrophages	Fibroblast proliferation, chemotaxis & collagen metabolism; angiogenesis; immunomodulation
Transforming growth factor-α	TGF-α	Platelets, keratinocytes	Keratinocyte proliferation & migration
Epidermal growth factor	EGF	Platelets	Keratinocyte proliferation & migration
Interleukins	IL-1, IL-10	Leukocytes, keratinocytes	Fibroblast proliferation; proinflammatory (IL-1); anti-inflammatory (IL-10)
Tumor necrosis factor	TNF	Leukocytes, keratinocytes	Promotes inflammation
Connective tissue growth factor	CTGF	Fibroblasts, endothelial cells	Fibroblast proliferation, chemotaxis & collagen metabolism
Fibroblast growth factor	FGF	Keratinocytes, macrophages	Fibroblast & epithelial cell proliferation; matrix deposition, wound contraction; angiogenesis
Keratinocyte growth factor	KGF	Fibroblasts	Keratinocyte proliferation
Insulin-like growth factor 1	IGF-1	Fibroblasts	Keratinocyte proliferation & differentiation
Hepatocyte growth factor	HGF	Fibroblasts, macrophages	Keratinocyte proliferation
Vascular endothelial growth factor	VEGF	Platelets, keratinocytes, macrophages, neutrophils	Angiogenesis; vascular permeability; macrophage chemotaxis



WOUND HEALING RESPONSE – CELLULAR AND MOLECULAR EVENTS
Tissue-resident epidermal stem cells (epidermal and hair follicle)
Interfollicular
Bulge
Injury
Fibroblast
Platelet activation and fibrin clot formation
FGF
PDGF
IL-1
KGF
HGF
Platelets
Repair of damaged tissues
Scar
Protease–protease inhibitor balance restored
Remodeling (maturation) phase
Wound contraction and re-epithelialization
Collagen fibers
Bone marrow-derived stem cells
Circulation
T cell
Mast cell
Neutrophil
Monocyte→tissue macrophage
MMPs
TIMPs
VEGF
EGF
IL-10
IL-6
IL-13
TNF
TGF-β
Inflammatory phase (and migration)
Resolution of inflammation
PDGF
TGF-β
Myofibroblast differentiation
Angiogenesis
VEGF
FGF, CTGF
Proliferative phase
Persistent inflammation
Chronic wound
Keratinocyte migration ↓
Hyperproliferation
Clot
Bacterial colonization
Biofilm formation
Fibrosis
Inflammation ↑
Protease activity ↑
Protease inhibitors ↓
Angiogenesis ↓
Growth factors ↓
Senescence (e.g. fibroblasts)

Fig. 141.3 Wound healing response – cellular and molecular events. Leakage of blood constituents and vasoactive factors into the wound site leads to hemostasis. The fibrin clot consists of platelets as well as extracellular matrix molecules (e.g. fibronectin, vitronectin) which facilitate the infiltration and migration of cells. Influx of different subsets of leukocytes characterizes the *inflammatory phase* of repair. These cells release angiogenic growth factors and proinflammatory cytokines, which then induce the recruitment of additional inflammatory cells, fibroblasts, and endothelial cells into the wound site. During the *proliferative phase* in which there is formation of granulation tissue, activated macrophages, endothelial cells, and fibroblasts form a functional unit essential for efficient blood vessel growth. Transformation of granulation tissue and provisional wound matrix into scar tissue occurs during the *remodeling phase* and this is characterized by a decrease in inflammation and regression of vascular structures. Lastly, stem cells that are intrinsic to the skin (e.g. epidermal stem cells) as well as circulating stem cells contribute to the repair response. If the inflammatory response persists, a disturbed healing response may ensue leading to chronic skin ulceration. The microenvironment of chronic ulcers is characterized by multiple hostile factors including increased protease activity, proinflammatory factors, senescent cells, fibrosis, and impaired cellular migration. CTGF, connective tissue growth factor; EGF, epidermal growth factor; FGF, fibroblast growth factor; HGF, hepatocyte growth factor; IL, interleukin; KGF, keratinocyte growth factor; MMPs, matrix metalloproteinases; PDGF, platelet-derived growth factor; TGF, transforming growth factor; TIMPs, tissue inhibitors of metalloproteinases; TNF, tumor necrosis factor; VEGF, vascular endothelial growth factor. *Adapted from Wynn TA. Common and unique mechanisms regulate fibrosis in various fibroproliferative diseases. J Clin Invest. 2007;117:524–9.*

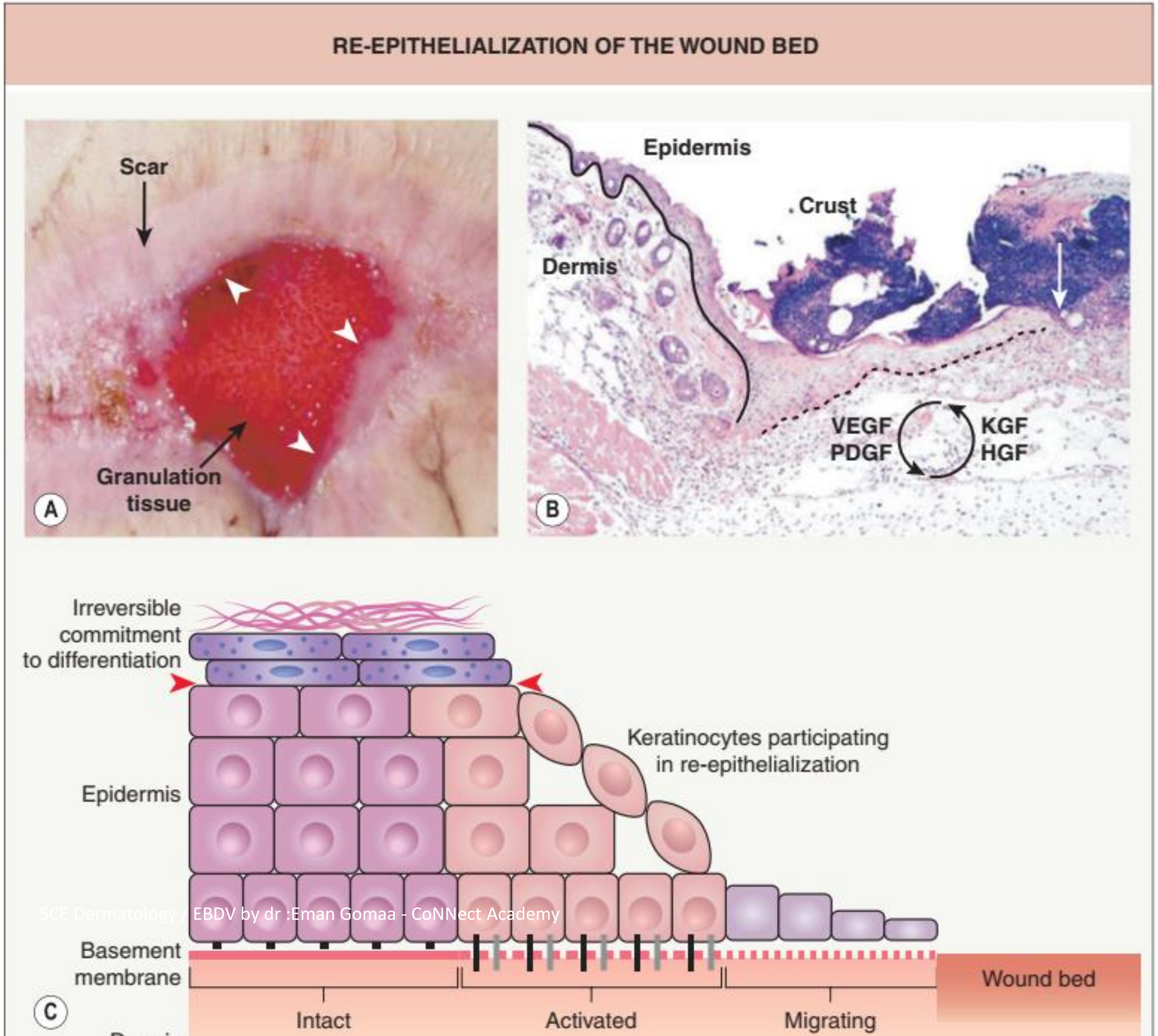
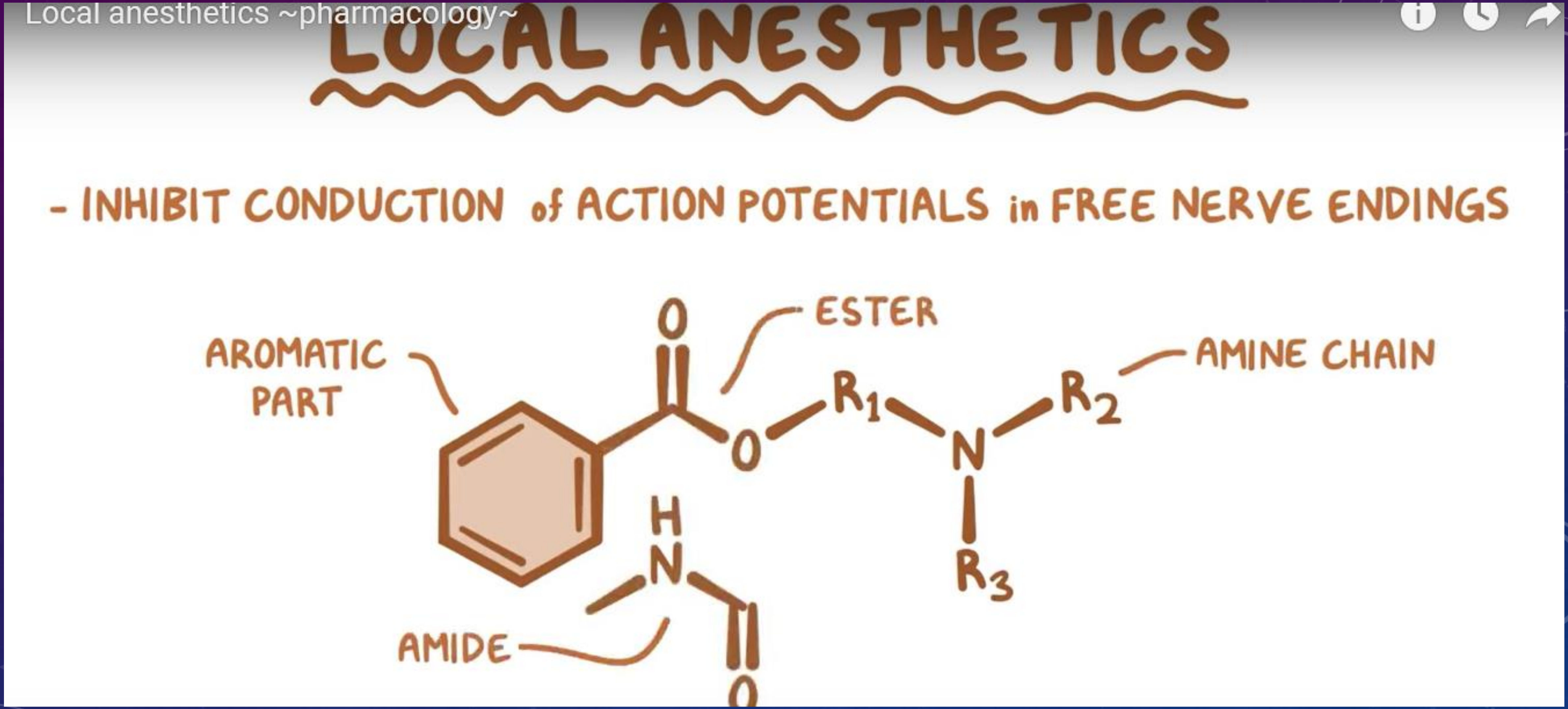
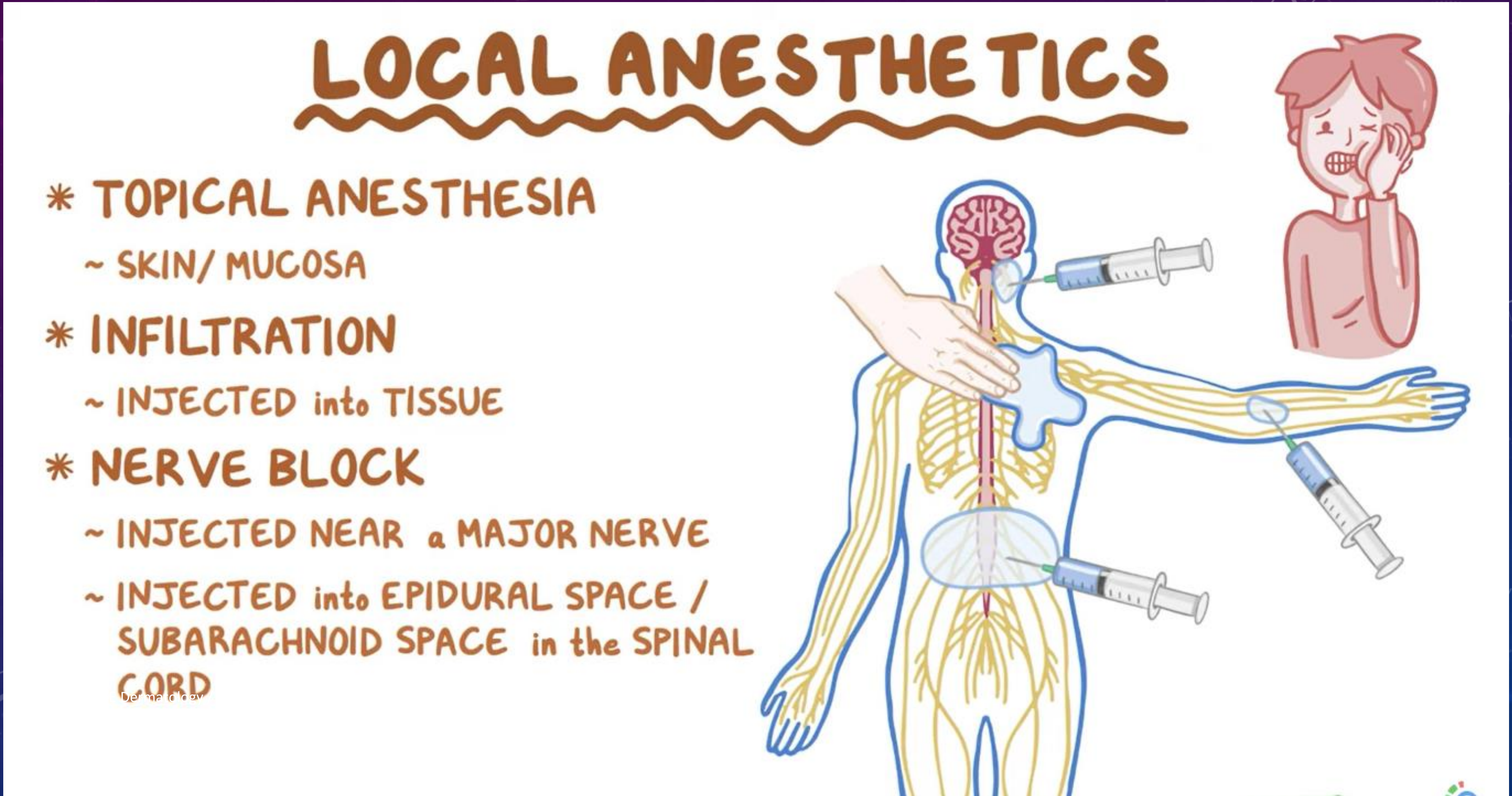
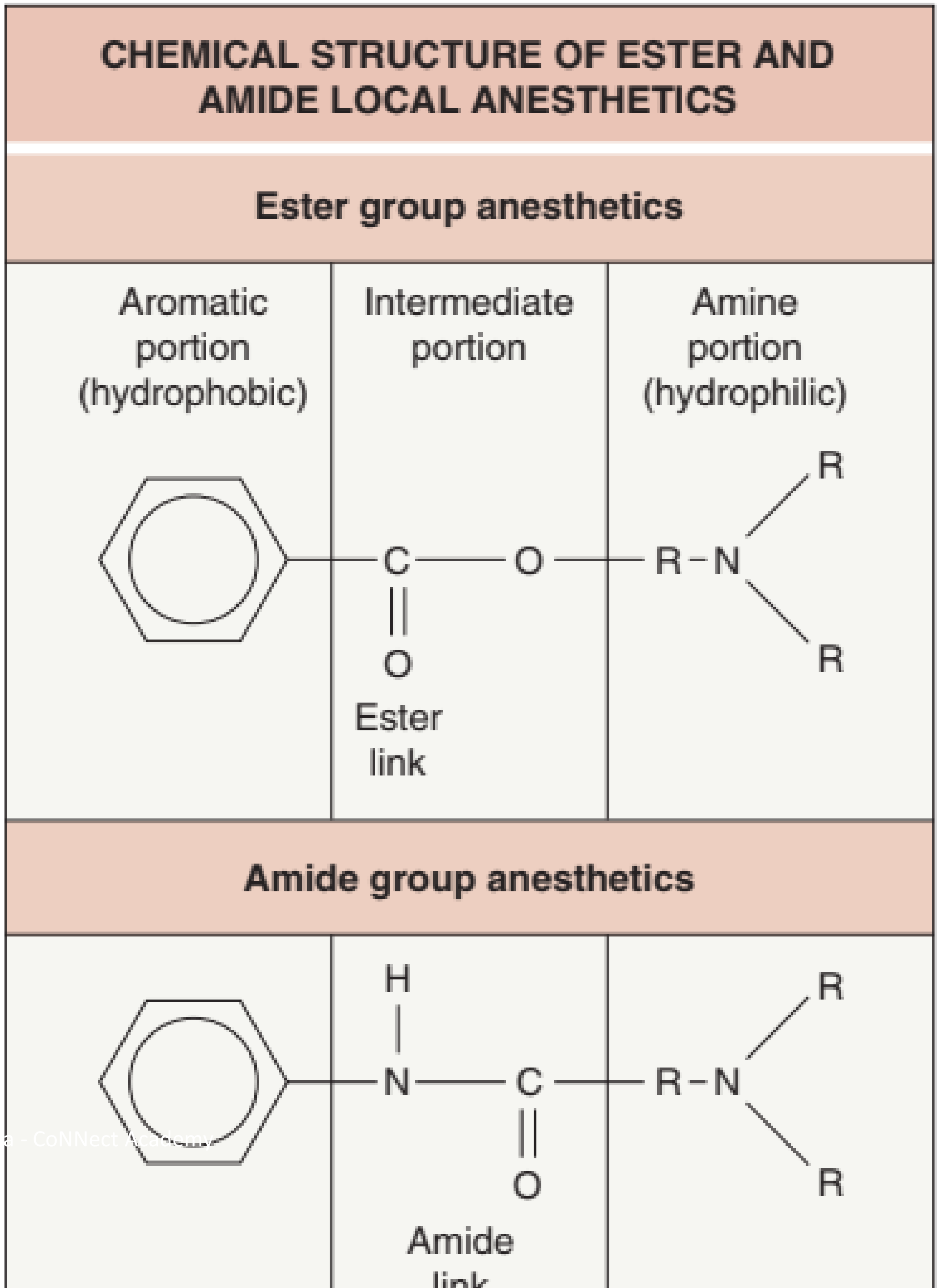


Fig. 141.4 Re-epithelialization of the wound bed. **A** Clinical appearance of a re-epithelializing wound; arrowheads indicate the epithelial wound edge. **B** Photomicrograph of the wound edge demonstrating the migration of keratinocytes over the wound bed; this process is in part regulated by growth factors synthesized by fibroblasts such as keratinocyte growth factor (KGF) and hepatocyte growth factor (HGF). The white arrow indicates the tip of the epithelial tongue, the black dashed line indicates activated and migrating epithelium, and the black solid line indicates intact epithelium. **C** Epithelial cells present at the wound edge undergo marked morphologic and functional alterations. They lose their contacts with neighboring cells and the underlying extracellular matrix, then proliferate and migrate toward the center of the wound. The red arrowheads demarcate the terminally differentiated cells that are irreversibly committed.

DIFFERENT TYPES OF MATRIX METALLOPROTEINASES INVOLVED IN WOUND REPAIR	
Enzyme	MMP number
Collagenase-1	MMP-1
Collagenase-2	MMP-8
Collagenase-3	MMP-13
Collagenase-4	MMP-18
Stromelysin-1	MMP-3
Stromelysin-2	MMP-10
Stromelysin-3	MMP-11
Gelatinase-A	MMP-2
Gelatinase-B	MMP-9
Matrilysin	MMP-7
Macrophage metalloproteinase (elastase)	MMP-12

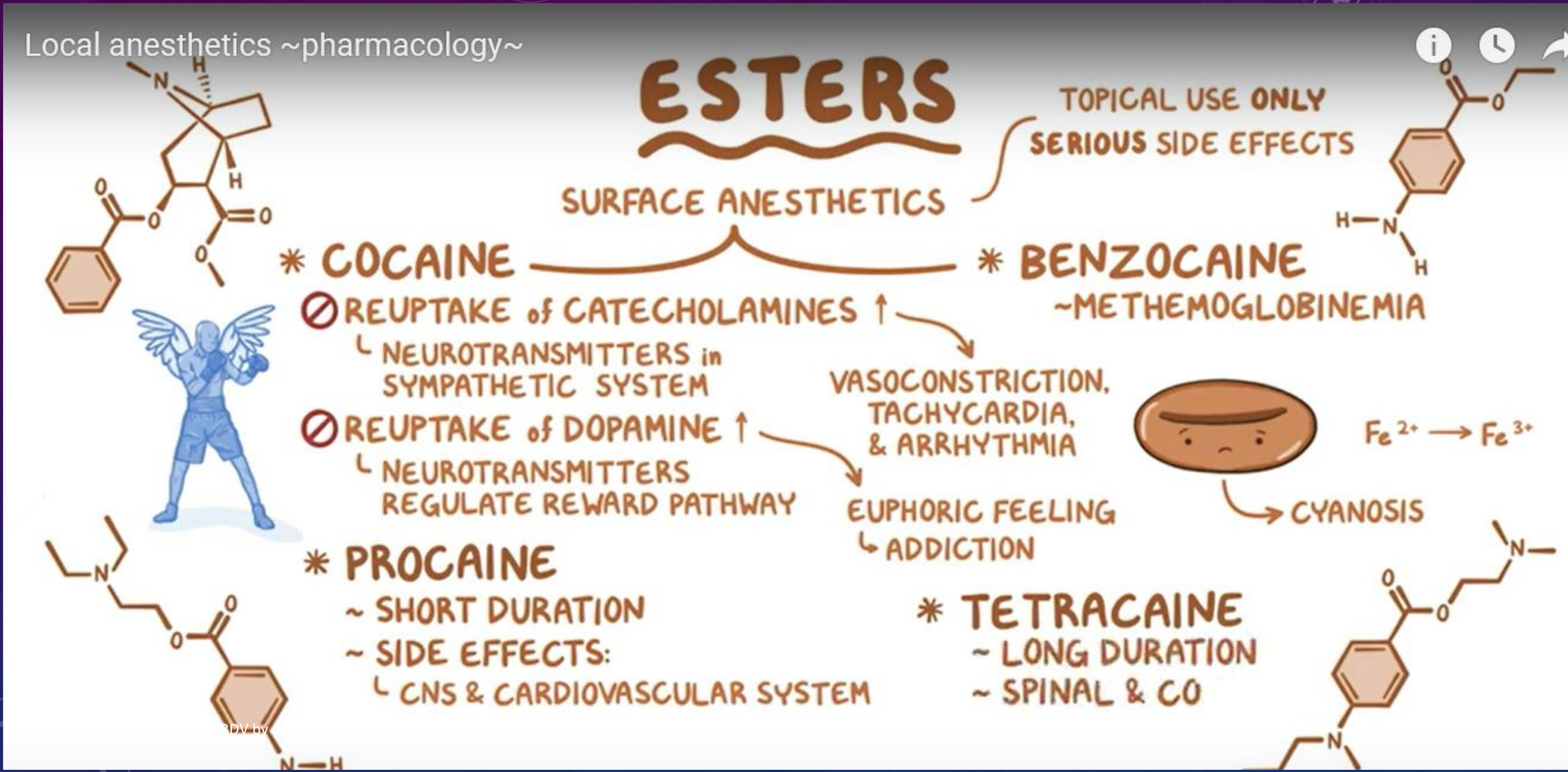

Local anesthetics ~pharmacology~
LOCAL ANESTHETICS
- INHIBIT CONDUCTION of ACTION POTENTIALS in FREE NERVE ENDINGS
AROMATIC PART
ESTER
AMINE CHAIN
O
O
R_1
N
R_2
R_3
H
N
AMIDE
O

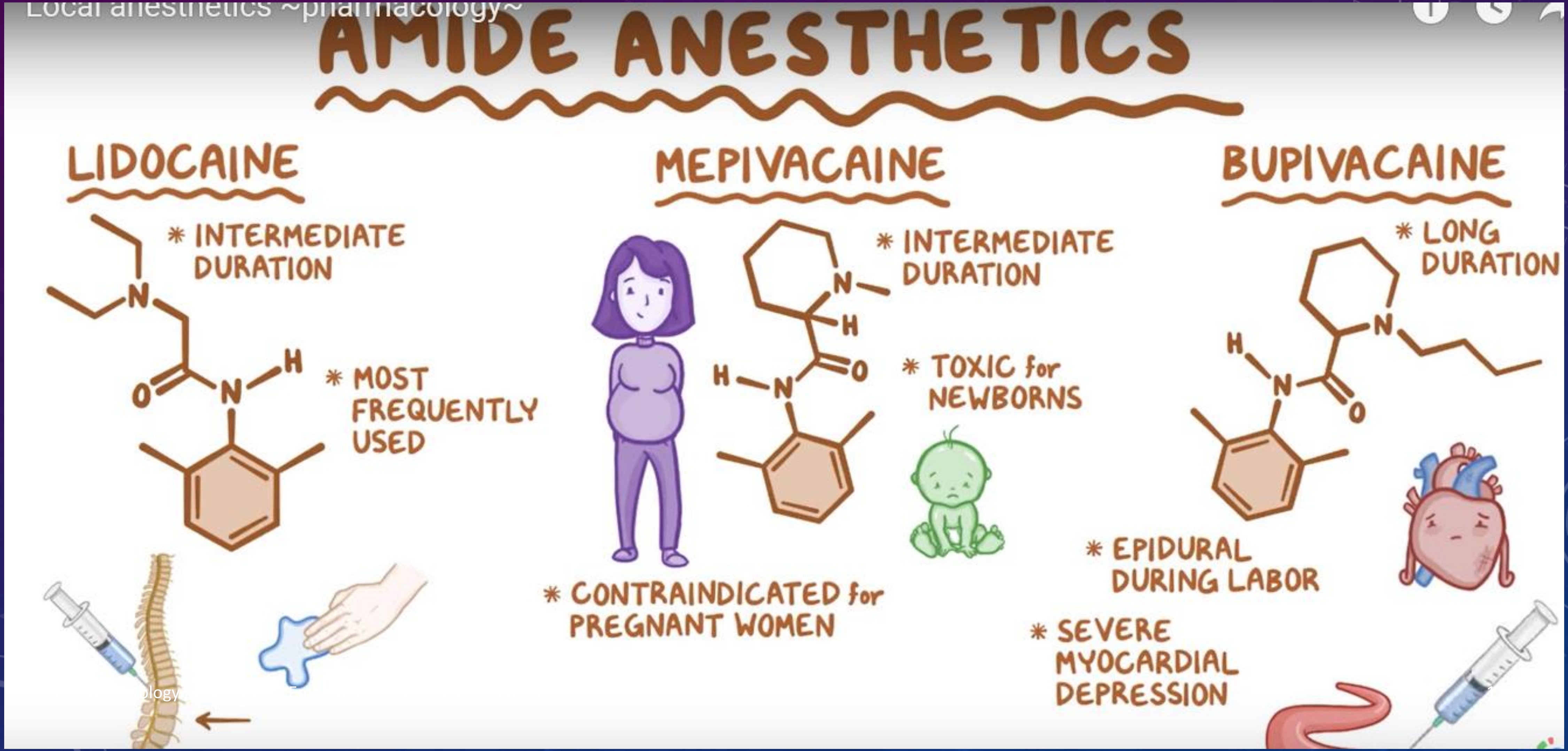

LOCAL ANESTHETICS
* TOPICAL ANESTHESIA
~ SKIN/ MUCOSA
* INFILTRATION
~ INJECTED into TISSUE
* NERVE BLOCK
~ INJECTED NEAR a MAJOR NERVE
~ INJECTED into EPIDURAL SPACE / SUBARACHNOID SPACE in the SPINAL CORD


CHEMICAL STRUCTURE OF ESTER AND AMIDE LOCAL ANESTHETICS
Ester group anesthetics
Aromatic portion (hydrophobic)
Intermediate portion
Amine portion (hydrophilic)
C
O
O
R
R–N
R
Ester link
Amide group anesthetics
H
N
C
O
R
R–N
R
Amide

LOCAL ANESTHETICS FOR INFILTRATIVE AND NERVE BLOCK ANESTHESIA					
Generic name	Trade name®	Onset (min)	Duration plain (h)	Maximum dose plain (mg for 70 kg person)	Maximum dose with epinephrine (mg for 70 kg person)
Amides					
Articaine	Septocaine	2–4	0.5–2	350	500
Bupivacaine hydrochloride	Marcaine	5–8	2–4	175	225
Etidocaine	Duranest	3–5	3–5	300	400
Levobupivacaine hydrochloride	Chirocaine	2–10	2–4	150	Not available
Lidocaine	Xylocaine	Rapid	0.5–2	350	500 (3500 dilute)
Mepivacaine	Carbocaine	3–20	0.5–2	300	500
Prilocaine hydrochloride	Citanest	5–6	0.5–2	400	600
Ropivacaine*	Naropin	1–15	2–6	200	Not available
Esters					
Chloroprocaine hydrochloride	Nesacaine	5–6	0.5–2	800	1000
Procaine	Novocaine	5	1–1.5	500	600
Tetracaine	Pontocaine	7	2–3	100	Not available

*Addition of epinephrine has no effect on onset or duration of action of ropivacaine.


Local anesthetics ~pharmacology~
ESTERS
TOPICAL USE ONLY
SERIOUS SIDE EFFECTS
SURFACE ANESTHETICS
* COCAINE
REUPTAKE of CATECHOLAMINES ↑
NEUROTRANSMITTERS in SYMPATHETIC SYSTEM
REUPTAKE of DOPAMINE ↑
NEUROTRANSMITTERS REGULATE REWARD PATHWAY
VASOCONSTRICTION, TACHYCARDIA, & ARRHYTHMIA
EUPHORIC FEELING
ADDICTION
* BENZOCAINE
~METHEMOGLOBINEMIA
$Fe^{2+} \rightarrow Fe^{3+}$
CYANOSIS
* PROCAINE
~ SHORT DURATION
~ SIDE EFFECTS:
CNS & CARDIOVASCULAR SYSTEM
* TETRACAINE
~ LONG DURATION
~ SPINAL & CO



Local anesthetics ~pharmacology~
AMIDE ANESTHETICS
LIDOCAINE
* INTERMEDIATE DURATION
* MOST FREQUENTLY USED
MEPIVACAINE
* INTERMEDIATE DURATION
* TOXIC for NEWBORNS
* CONTRAINDICATED for PREGNANT WOMEN
BUPIVACAINE
* LONG DURATION
* EPIDURAL DURING LABOR
* SEVERE MYOCARDIAL DEPRESSION

DIFFERENTIAL DIAGNOSIS OF LOCAL ANESTHETIC SYSTEMIC REACTIONS				
Diagnosis	**Pulse rate**	**Blood pressure**	**Signs and symptoms**	**Emergency management**
Vasovagal reaction	Low	Low	Excess parasympathetic tone; diaphoresis, hyperventilation, nausea	Trendelenburg, cold compress, reassurance
Epinephrine reaction	High	High	Excess α- and β-adrenergic receptor stimulation; palpitations	Reassurance (usually resolves within minutes), phentolamine, propranolol
Anaphylactic reaction	High	Low	Peripheral vasodilation with reactive tachycardia; stridor, bronchospasm, urticaria, angioedema	Epinephrine 1:1000 0.3 ml subcutaneous, antihistamines, corticosteroids, fluids, oxygen, airway maintenance
Lidocaine overdose				
1–6 mcg/ml	Normal	Normal	Circumoral and digital paresthesias, restlessness, drowsiness, euphoria, lightheadedness	Observation
6–9 mcg/ml	Normal	Normal	Nausea, vomiting, muscle twitching, tremors, blurred vision, tinnitus, dysgeusia (e.g. metallic taste), confusion, excitement, psychosis	Diazepam, airway maintenance
9–12 mcg/ml	Low	Low	Seizures, cardiopulmonary depression, arrhythmia	Respiratory support
>12 mcg/ml	None	None	Coma, cardiopulmonary arrest	Cardiopulmonary resuscitation and life support

TOPICAL ANESTHETICS FOR MUCOUS MEMBRANE AND INTACT SKIN ANESTHESIA				
Generic name	Trade name®	Concentration (%)	Type	Primary use
Benzocaine	Americaine otic	20	Ester	Tympanic membrane
Benzocaine	Hurricaine	20	Ester	Mucous membranes
Benzocaine/tetracaine/butamben (butyl aminobenzoate)	Cetacaine, Exactacain		Esters	Mucous membranes
Benzocaine/lidocaine/tetracaine	compounded	Various	Ester/amide/ester	Intact skin
Cocaine hydrochloride		2–10	Ester	Nasal mucosa
Dibucaine (cinchocaine)	Nupercaine	1	Amide	Mucous membranes
Lidocaine	LMX	4–5	Amide	Intact skin (4%); anorectal (5%)
Lidocaine	Topicaine	4	Amide	Intact skin
Lidocaine	Xylocaine	2–5	Amide	Mucous membranes
Lidocaine in acid mantle cream	compounded	30–40	Amide	Intact skin
Lidocaine/prilocaine eutectic mix [cream or disc]	EMLA	2.5/2.5	Amides	Intact skin
Lidocaine/prilocaine mix	Betacaine LA	Proprietary	Amides	Intact skin
Lidocaine/tetracaine peel	Pliaglis	7/7	Amide/ester	Intact skin
Lidocaine/tetracaine patch	Synera	7/7	Amide/ester	Intact skin
Proparacaine (proxymetacaine)	Alcaine	0.5	Ester	Conjunctiva
Tetracaine gel	Tetracaine gel	4	Ester	Intact skin
Tetracaine	Pontocaine	0.5	Ester	Conjunctiva

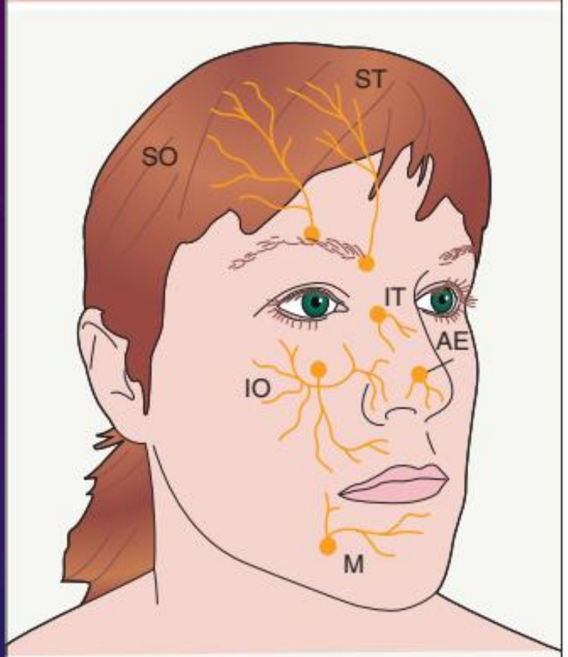
MAXIMUM RECOMMENDED EMLA® APPLICATION IN CHILDREN				
Age	Weight (kg)	Maximum total dose of cream (g)	Maximum area (cm^2)	Maximum time (hours)
<3 months	<5	1	10	1
3–12 months	5–10	2	20	4
1–6 years	10–20	10	100	4
7–12 years	>20	20	200	4

MAXIMUM RECOMMENDED LIDOCAINE 4% LIPOSOMAL CREAM (LMX4®) APPLICATION IN CHILDREN

Age	Weight (kg)	Maximum area	Maximum time (hours)
<12 months	<10	100 cm^2 (4 × 4 in)	1
1–6 years	10–20	200 cm^2 (5.5 × 5.5 in)	2
7–12 years	>20	600 cm^2 (9 × 9 in)	2*

*0.072 mg 4% lidocaine absorption/cm^2/h × 600 cm^2 × 2 h = 86 mg (≤4 mg/kg).

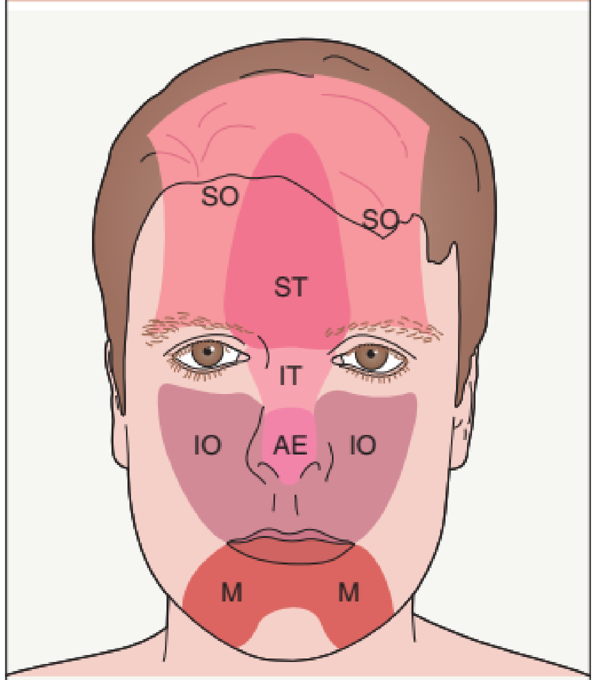
SKULL EXIT POINTS OF CENTRAL FACE SENSORY NERVES



AE External nasal branch of the anterior ethmoidal nerve (V1)
IO Infraorbital nerve (V2)
IT Infratrochlear nerve (V1)
M Mental nerve (V3)
SO Supraorbital nerve (V1)
ST Supratrochlear nerve (V1)

Fig. 143.3 Skull exit points of central face sensory nerves.

SENSORY INNERVATION OF THE CENTRAL FACE



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M Mental nerve (V3)
SO Supraorbital nerve (V1)
ST Supratrochlear nerve (V1)

Fig. 143.2 Sensory innervation of the central face.

Fig. 143.4 Locations of needle insertions for central face nerve blocks. From superior to inferior, the supraorbital, infraorbital and mental nerves are marked with an "x" and the supratrochlear, infratrochlear and external nasal nerves are marked with a rectangle.

Fig. 143.5 Intraoral route for infraorbital nerve block. The needle is inserted cephalad between the premolar (bicuspid) teeth in the midpupillary line.

WOUND CLOSURE MATERIALS AND INSTRUMENTS

COMMONLY USED ABSORBABLE SUTURES

Suture	Configuration	Tensile strength	Ease of handling	Knot security*	Tissue reactivity	Uses
Surgical gut (plain)	Virtually monofilament	Poor at 7–10 days	Fair	Poor	Moderate	Rarely used today in skin
Surgical gut (chromic)	Virtually monofilament	Poor at 21–28 days	Poor	Poor	Less than plain	Skin grafts, surface sutures for mucosae
Surgical gut (fast-absorbing)	Virtually monofilament	50% at 3–5 days	Fair	Poor	Low	Skin grafts, epidermal closure
Polyglycolic acid (Dexon®)	Braided†	20% at 21 days	Good	Good	Low	Subcutaneous closure, vessel ligature
Polyglactin 910 (Vicryl®, Polysorb®)	Braided†	75% at 14 days 50% at 21 days 25% at 28 days	Good	Good	Low	Subcutaneous closure, vessel ligature
Polydioxanone (PDS II®)	Monofilament	wks / 4/0 and smaller / 3/0 and larger 2 / 60% / 80% 4 / 40% / 70% 6 / 35% / 60%	Poor	Moderate	Low	Subcutaneous closure (high-tension areas)
Polyglyconate (Maxon®)	Monofilament	80% at 14 days 60% at 28 days	Fair	Good	Low	Subcutaneous closure (high-tension areas)
Poliglecaprone 25 (Monocryl®)	Monofilament	50–60% at 7 days 20–30% at 14 days 0% at 21 days	Fair	Moderate	Minimal	Subcutaneous closure (avoid if high tension site)
Glycomer 631 (Biosyn®)	Monofilament	75% at 14 days 40% at 21 days	Fair	Moderate	Minimal	Subcutaneous closure; similar to poliglecaprone 25 plus an additional week of tensile strength
Polyglytone 6211 (Caprosyn®)	Monofilament	20% at 10 days	Fair	Moderate	Minimal	Subcutaneous closure: more rapid absorption than poliglecaprone 25

*Directly proportional to the friction coefficient and indirectly proportional to memory.
†Multifilament

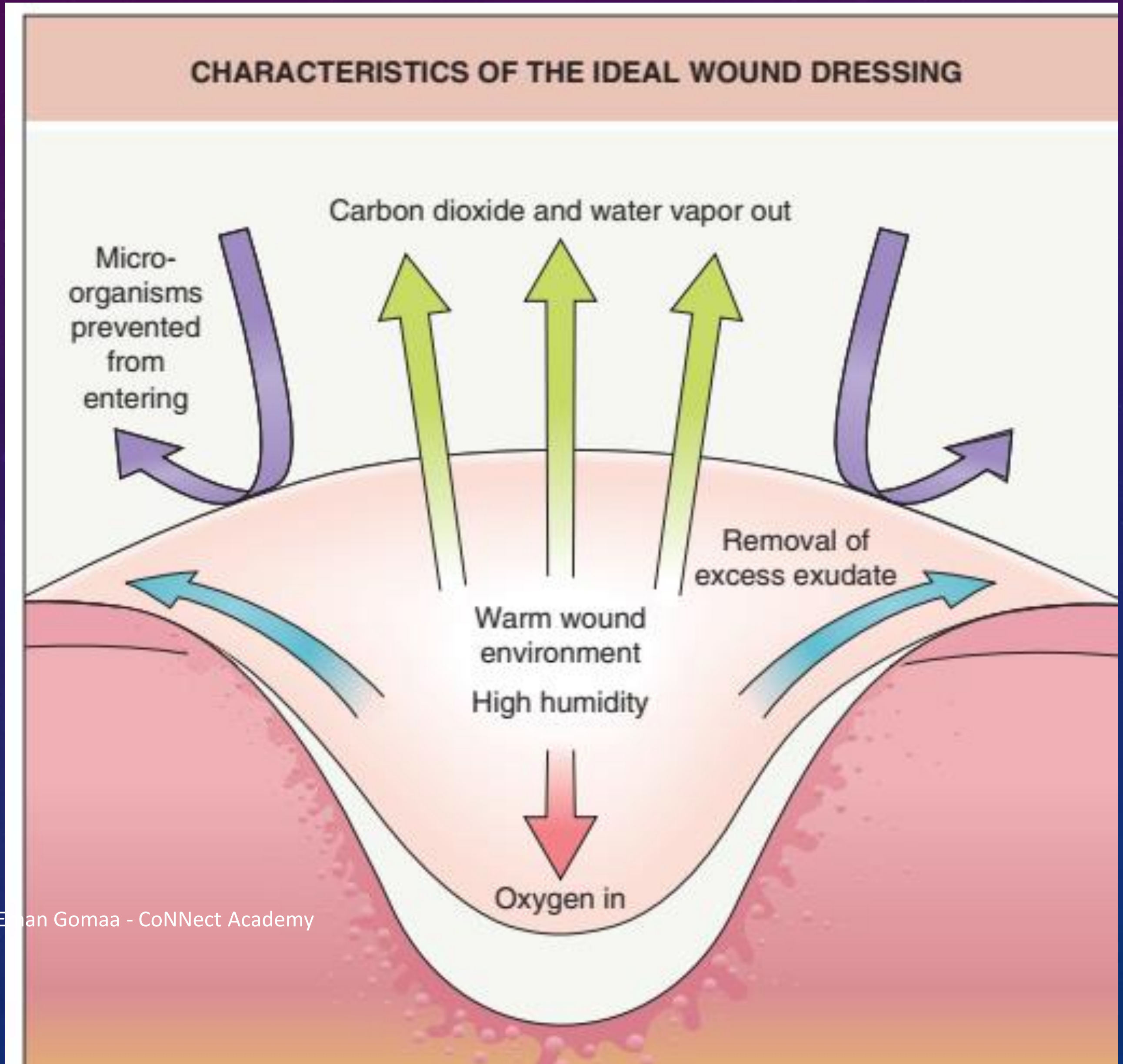
COMMONLY USED NON-ABSORBABLE SUTURES						
Suture	Configuration	Tensile strength	Ease of handling	Knot security*	Tissue reactivity	Uses
Silk	Braided†	No decrease in 365 days	Gold standard	Good	Moderate	Mucosal surfaces
Nylon:						
Ethilon®	Monofilament	Decreases 20% per year	Good to fair	Poor	Low	Epidermal closure
Dermalon®	Monofilament	Good	Good to fair	Poor		
Surgilon®	Braided†	Good	Good	Fair		
Nurolon®	Braided†	Good	Good	Fair		
Polypropylene (Prolene®, Surgilene®, Surgipro®)	Monofilament	Extended	Good to fair	Poor	Minimal	Running subcuticular suture
Polyester (Dacron®, Mersilene®, Ethibond®)	Braided†	Indefinite	Very good	Good (coating decreases)	Minimal	Mucosal surfaces
Polybutester (Novafil®)	Monofilament	Extended	Good to fair	Poor	Low	Epidermal closure

COMMONLY UTILIZED SUTURES BY SITE			
Site	**Deep suture**	**Surface suture**	**Other suture**
Face	4–0/5–0 polyglactin 910 or poliglecaprone 25	(1) 5–0/6–0 nylon or polypropylene (2) 6–0 fast-absorbing gut or partially hydrolyzed polyglactin 910	
Neck and distal extremities	4–0 polyglactin 910 or poliglecaprone 25	(1) 4–0/5–0 nylon or polypropylene (2) 4–0/5–0 poliglecaprone 25 running subcuticular	
Trunk, proximal extremities, and scalp	3–0/4–0 polyglactin 910 (any) or poliglecaprone 25 (low tension) or polydiaxonone (high tension)	(1) 3–0/4–0 nylon or polypropylene (2) 4–0/5–0 poliglecaprone 25 running subcuticular	
Mucosa	None	(1) 5–0 silk (2) 5–0 polyester or 5–0/6–0 polyglactin 910	
Vascular ligature			3–0/4–0 polyglactin 910

Fig. 144.7 Facial surgical instruments. Authors' tray for facial surgery/repair includes (from left to right) two curved Mosquito hemostats, suture scissors, two double-pronged skin hooks, delicate Webster needle holder, Supercut® iris scissors, delicate Supercut® Shea undermining scissors, 0.9 mm Castroviejo forceps, and Siegel blade handle with #15 blade. Two towel clamps are above the tissue scissors.

STERILIZATION METHODS		
Method	**Advantages**	**Disadvantages**
Steam autoclave	Most popular in office; easiest; safest	Must use 20–30 min at 2 atm pressure and 121°C; corrosive; may dull sharp instruments
Chemiclave	Lower humidity than steam; less dulling of sharp instruments; instruments are drier	Special chemical needed (mixture of formaldehyde, methyl ethyl ketone, acetone and alcohols)
Dry heat (oven)	Inexpensive; no corrosion or dulling	High temperature, longer time (1 h at 171°C; 6 h at 121°C); cannot use cloth, paper or plastic
Gas sterilization	Good for large volumes (mostly used in hospitals)	Expensive equipment; prolonged times (1 day for paper, 7 days for polyvinyl chloride); toxic and mutagenic gas
Cold sterilization: glutaraldehyde or ortho-phthalaldehyde solutions	Simple; inexpensive; used for heat-sensitive equipment (e.g. endoscopes)	Irritating to skin (glutaraldehyde); not recommended as only method; not always effective against bacterial spores (ortho-phthalaldehyde has superior sporicidal activity)

❑ DRESSINGS

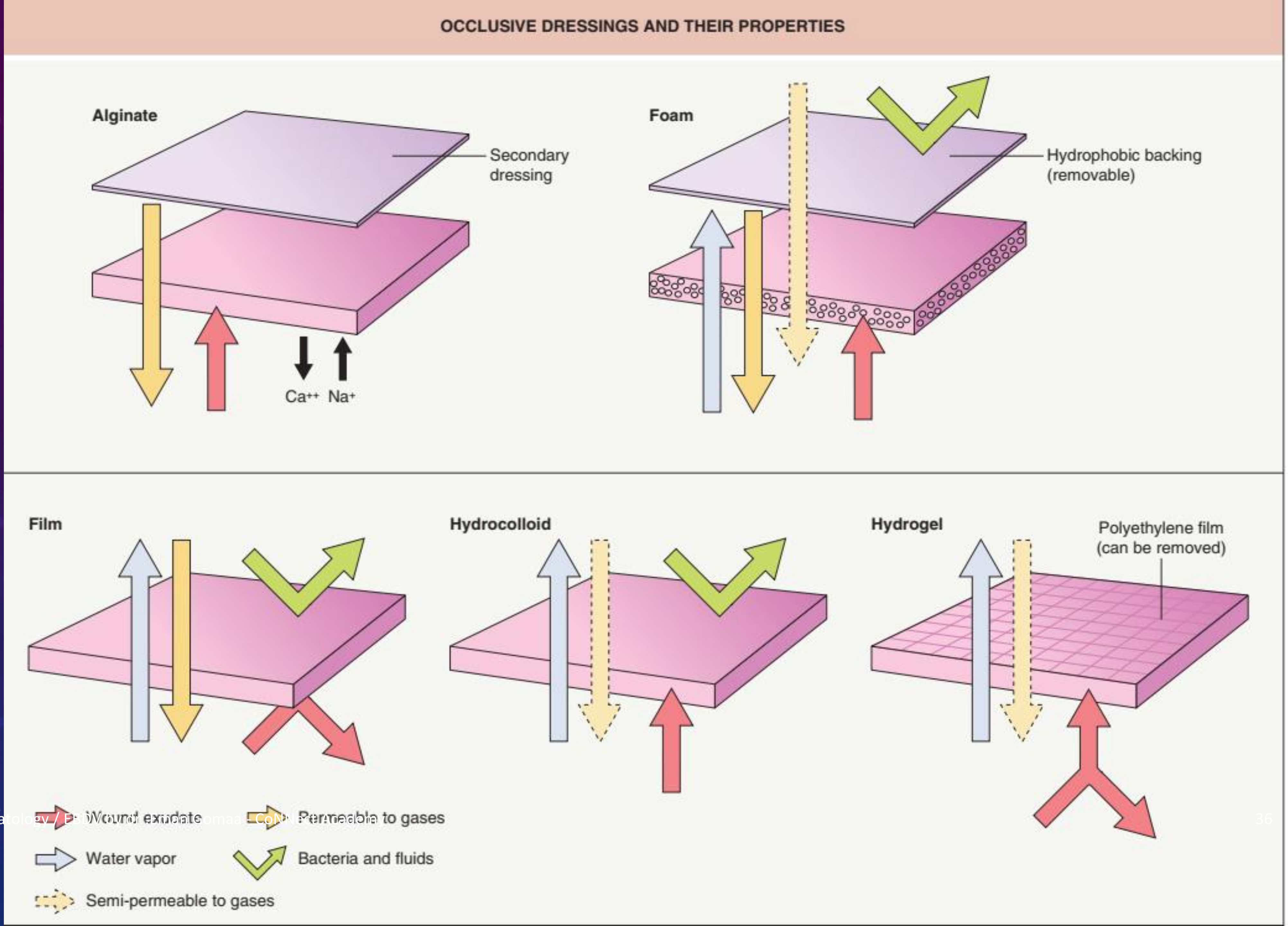

CHARACTERISTICS OF THE IDEAL WOUND DRESSING
Carbon dioxide and water vapor out
Micro-
organisms
prevented
from
entering
Removal of
excess exudate
Warm wound
environment
High humidity
Oxygen in

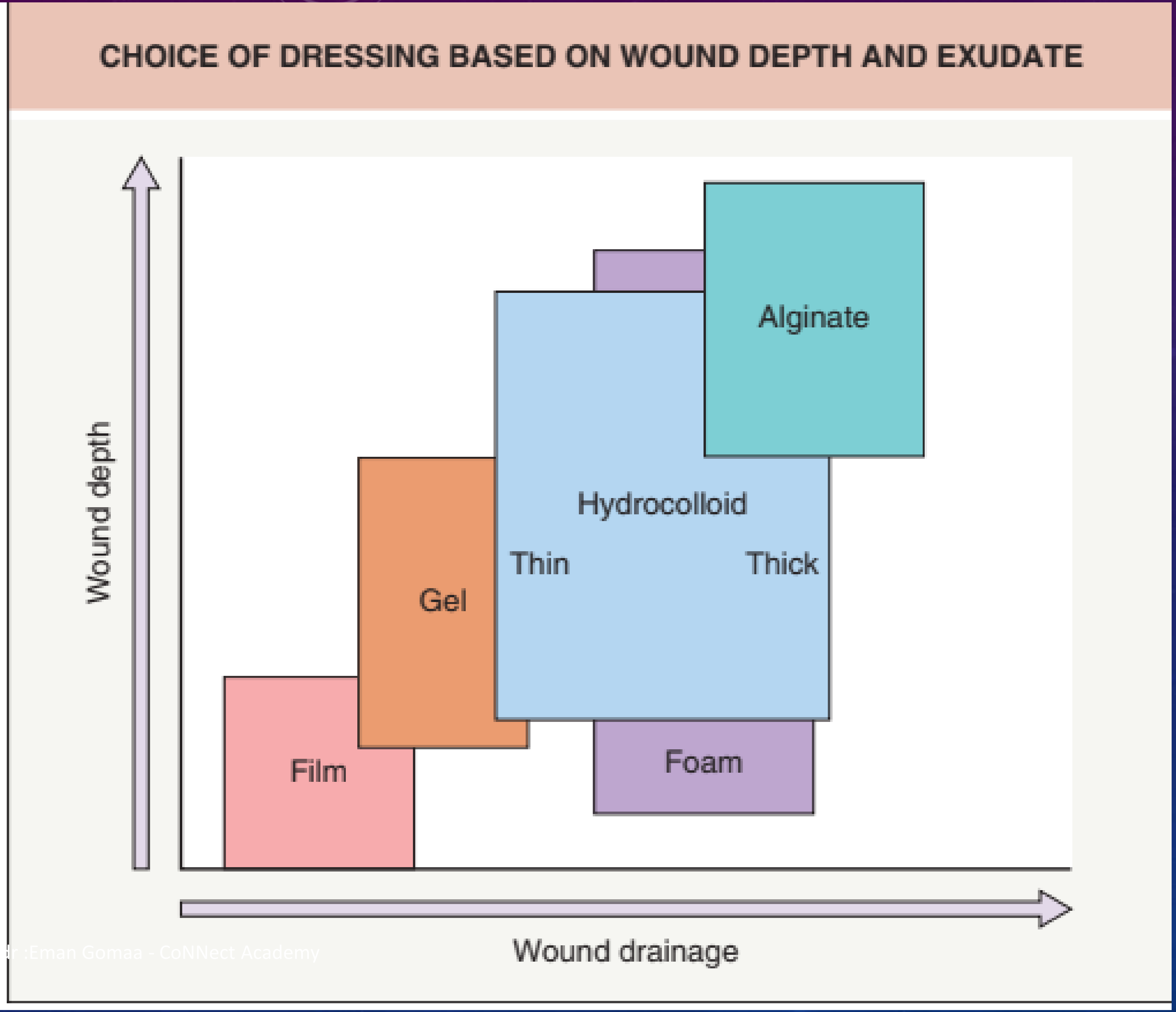
SELECTION OF APPROPRIATE DRESSINGS	
Wound factor	**Dressing**
Exudate	
• Provides moisture	Hydrogel
• Maintains moisture	Film, hydrocolloid
• Absorbs moisture	Foam, alginate, gelling fibers
Pain	
• Reduces pain	Hydrogel, soft silicone, non-adherent
Bleeding	
• Contains hemostatic agents	Alginates
Superficial infection	
• Reduces number of microorganisms	Impregnated with silver, chlorhexidine, iodine, honey, or methylene blue plus gentian violet as antiseptics

CHARACTERISTICS OF AN IDEAL DRESSING – COMPARISON OF MOISTURE-RETENTIVE DRESSINGS					
	Comfort	Absorbency	Pain relief	Easy to re-apply	Debridement
Films			+		+
Foams	+	++	+	+	
Hydrogels	++	+	++	+	++
Alginates	++	++		+	+
Hydrocolloids	+	++	+	+	+

Fig. 145.2 Second intention healing. A Surgical wound immediately following Mohs micrographic surgery; a semi-occlusive dressing will be applied. **B** Three weeks following surgery. Note the abundance of pink–red granulation tissue. **C** Fully re-epithelialized wound six weeks following surgery. *Courtesy, Gregg M Menaker, MD.*

DRESSINGS THAT CONTAIN ANTISEPTICS				
Antimicrobial agent	**Examples of dressings**	**Coverage spectrum**	**Advantages**	**Disadvantages**
Silver	Acticoat® Flex 7 Silver, Actisorb® Silver 220, Algicell® Ag (alginate), Aquacel® Ag, Contreet® Biatain Ag Foam, ColActive® Plus Ag, Mepilex® Ag, PolyMem® Silver, SilvaSorb® (hydrogel), Silvercel® (alginate), Silverlon®, UrgoTul® Ag/Silver, UrgoTul® SSD	Broad-spectrum antibacterial, including MRSA and VRE	Release antibacterial levels of silver for 3–7 days Appear to decrease the levels of matrix metalloproteinases that are upregulated in nonhealing, chronic wounds Microbial resistance is rare Variety of products available for different wound situations Infrequent application required	Silver may stain tissues (localized argyria) Relatively expensive Slows acute wound healing
Chlorhexidine or polyhexamethylene biguanide (PHMB), a chlorhexidine derivative	Bactigras™ (chlorhexidine), Kendall™ AMD foam (PHMB), Kerlix™ AMD gauze (PHMB)	Broad-spectrum	Low tissue toxicity Different forms available, including foam, ribbons and gauze Comforting so recommended for painful wounds	May cause stinging and ICD Cytotoxicity – cornea, inner ear (in the setting of tympanic membrane rupture), and cartilage Systemic and local hypersensitivity reactions
Povidone-iodine* Cadexomer-iodine polymer*	Inadine®, Betadine® cream applied to gauze, Iodoflex™, Iodosorb®	Broad-spectrum antimicrobial – bacteria, fungi, viruses	Less irritating to skin than iodine alone Cadexomer-iodine polymer does not inhibit wound healing	May cause stinging ICD, ACD Povidone-iodine can inhibit wound healing
Honey	Activon Tulle, Medihoney® Calcium (alginate, hydrogel)	Broad-spectrum antimicrobial – bacteria, fungi, viruses	Low tissue toxicity Autolytic debridement	May cause maceration ACD to propolis
Methylene blue and gentian violet	Hydrofera Blue®	Broad-spectrum antibacterial, including MRSA and VRE, and anti-candidal	Less irritating Good absorptive capacity for cavity wounds	Bacteriostatic
Hypertonic saline	Mesalt®	Broad-spectrum	Less irritating to skin	


OCCLUSIVE DRESSINGS AND THEIR PROPERTIES
Alginate
Secondary dressing
Ca++ Na+
Foam
Hydrophobic backing (removable)
Film
Hydrocolloid
Hydrogel
Polyethylene film (can be removed)
Wound exudate
Permeable to gases
Water vapor
Bacteria and fluids
Semi-permeable to gases


CHOICE OF DRESSING BASED ON WOUND DEPTH AND EXUDATE
Wound depth
Alginate
Hydrocolloid
Thin
Thick
Gel
Film
Foam
Wound drainage

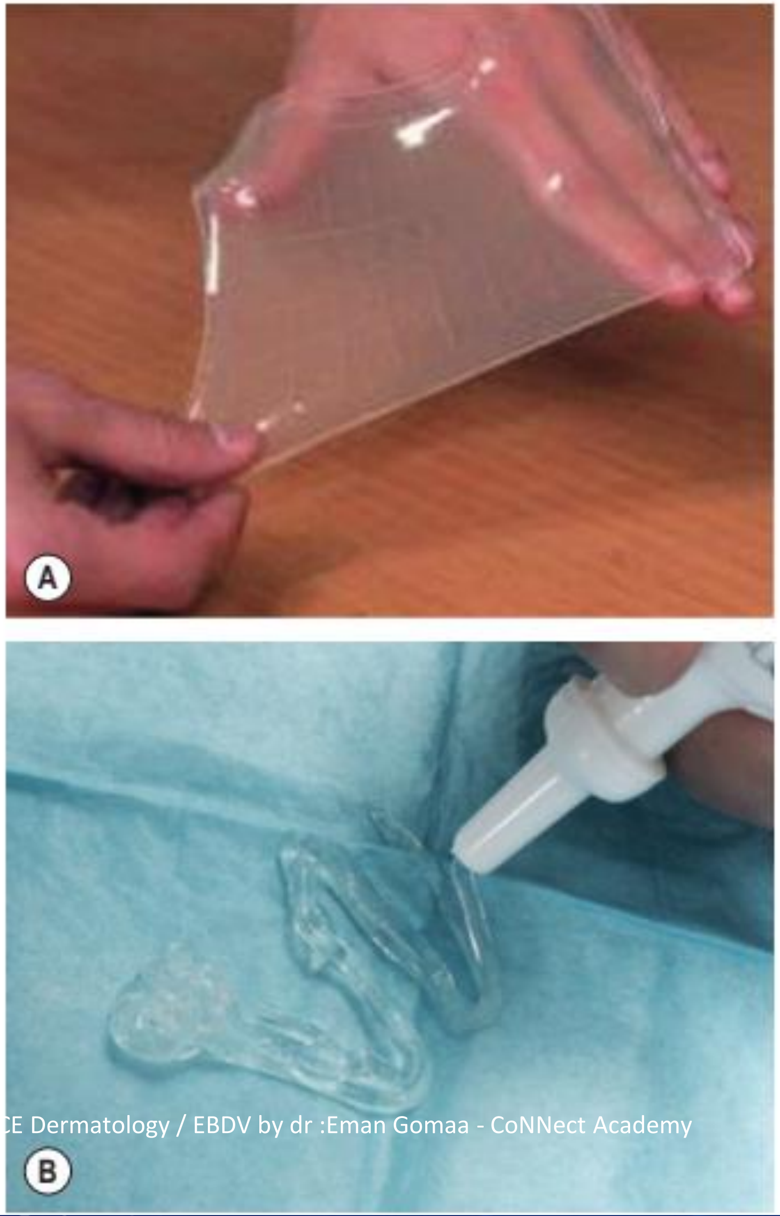


Fig. 145.7 Hydrogels.
A Hydrogel sheet.
B Premixed amorphous hydrogel.

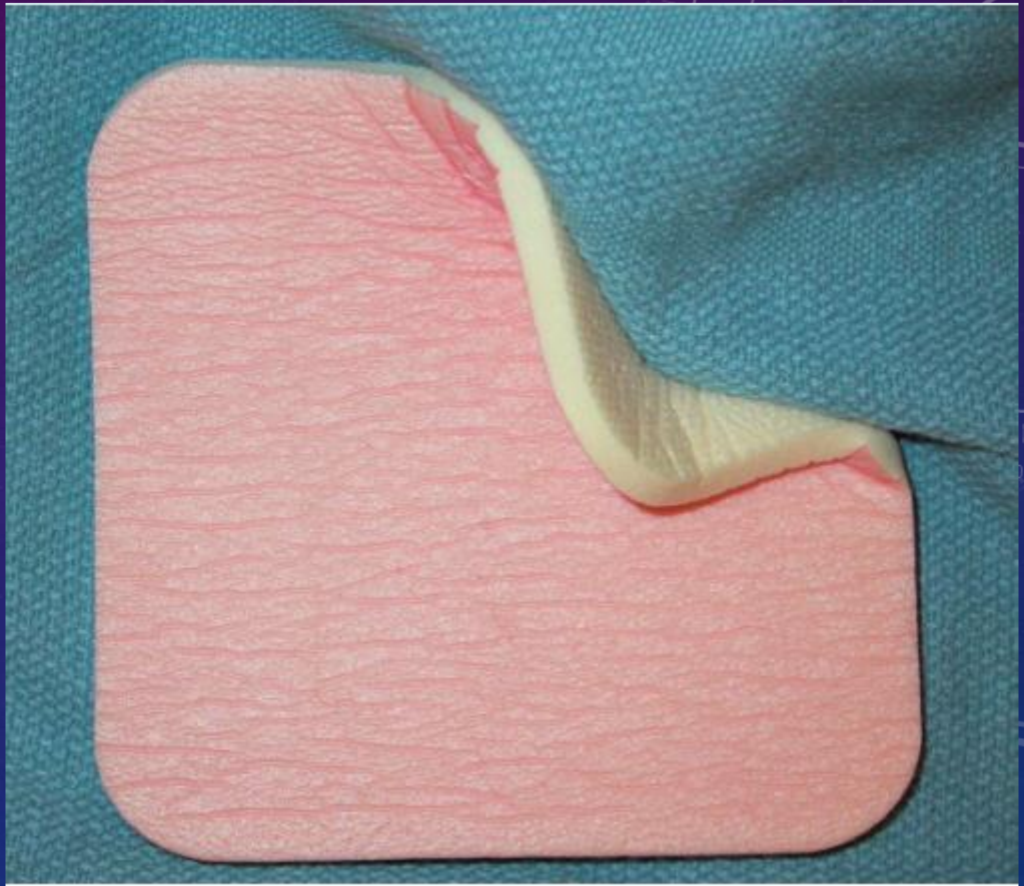


Fig. 145.6 Polymer foam dressing. It consists of hydrophilic foam with a hydrophobic backing (pink).

Fig. 145.8 Collagen–alginate complex dressing. Alginate dressings have hemostatic effects and can be used post

Fig. 145.9 Hydrocolloid dressing placed over a healing wound on the ankle. This example is DuoDERM®.

GROWTH FACTORS UNDER INVESTIGATION FOR TREATMENT OF CHRONIC ULCERS AND WOUNDS		
Growth factor	**Description**	**Studies**
Autologous platelet-rich fibrin matrix (PRFM)	Fibrinogen-containing plasma and platelets isolated from autologous blood form a fibrin membrane containing intact platelets with their growth factors, e.g. PDGF	e4
Platelet rich plasma (PRP)/Platelet gel (PG)	PRP is a supernatant of platelet-enriched plasma isolated from autologous blood PG is a combination of platelets with autologous thrombin	e5–e9
Granulocyte-macrophage colony-stimulating factor (GM-CSF)	Recombinant human hematopoietic growth factor	e10–e13
Basic fibroblast growth factor (bFGF)	Promotes angiogenesis and stimulates proliferation of keratinocytes and dermal fibroblasts	e14–e16
Vitronectin:growth factor complex (cVN:GF)	Complexes of IGF, IGF-binding protein-3, and EGF are bound to vitronectin, an ECM protein	e17,e18
Platelet-derived growth factor (PDGF)*	Released by platelets as well as macrophages, endothelial cells, fibroblasts and keratinocytes during the wound healing cascade (see Ch. 141)	e19,e20
Calcitonin gene-related peptide (CGRP) and vasoactive intestinal polypeptide (VIP)	CGRP is a neuropeptide, vasodilator and immunomodulator VIP is a vasodilator and immunomodulator	e21

DERMAL REPLACEMENT GRAFTS

	Description	Comments
Xenogeneic		
Biobrane®	Bilayer dressing: inner nylon mesh fabric coated with **porcine** type I collagen ("dermis") bonded to a semipermeable silicone film membrane ("epidermis")	Tissue ingrowth into the nylon fabric following binding of collagen to fibrin[e24] Two versions: (1) trilaminar nylon (better for clean, partial-thickness burn wounds or donor sites); and (2) monofilament nylon (better for meshed autografts) Approved for burns
EZ Derm™	**Porcine**-derived type I collagen attached to a gauze liner	Gauze liner is discarded upon placement[e25] Approved for partial-thickness skin loss injuries
Oasis®	Derived from porcine small intestine submucosa (mucosa and muscle removed) **Porcine** type I collagen, extracellular matrix (e.g. hyaluronic acid, fibronectin), and growth factors (FGF, TGF-β)	In study of 120 patients with venous leg ulcers, after 12 weeks of treatment, 55% of Oasis® + compression group healed vs 34% in compression alone group; at 6 months, 100% and 70% remained healed, respectively[e23] Approved for venous, diabetic and pressure ulcers, second-degree burns, graft donor sites, surgical wounds, and full-thickness skin injuries
Integra™	Bilayer dressing: **bovine** tendon collagen admixed with shark cartilage-derived, modified GAGs ("dermis") plus an outer silicone membrane	Debridement required prior to graft placement with sutures or staples Outer silicone cover removed after several weeks; in large wounds, replaced by ultrathin autograft (two-step process) In a controlled trial of matched-pair thermal injuries, as efficacious as other xenografts and synthetic skin substitutes[e26] Approved for life-threatening third-degree burns
Allogeneic		
Alloderm®*	Human cadaveric skin, chemically treated and purified to remove epidermis, all cellular components, and pathogens Cryopreserved (shelf life of 6 months)	Repopulation of host fibroblasts and endothelial cells seen within 10 days of grafting Approved for repair or replacement of damaged or inadequate integumental tissue, e.g. breast reconstruction
Dermagraft®	Neonatal foreskin fibroblasts are seeded onto a 3-D scaffold of biodegradable polyglactin mesh	Similar time to graft take and similar fibroblast function as TransCyte™ (see above) In 235 patients with diabetic ulcers, improved but not a significant difference in healing time compared to conventional therapy Approved for full-thickness diabetic foot ulcers and chronic ulcers that are deep but do not involve tendons, muscle or bone; previously HDE for ulcers in epidermolysis bullosa[e27]
Xenogeneic & allogeneic		
Transcyte™	Biobrane® (see above) plus neonatal foreskin fibroblasts	Requires ~2 weeks for graft take Fibroblasts initially secrete extracellular matrix (including collagen) and growth factors that stimulate granulation tissue and epithelialization but then die In a controlled trial of excised burns, easier to use and similar outcomes when compared to human cadaveric skin Approved for partial- and full-thickness burns

Fig. 145.12 Example of an engineered composite (xenogeneic and

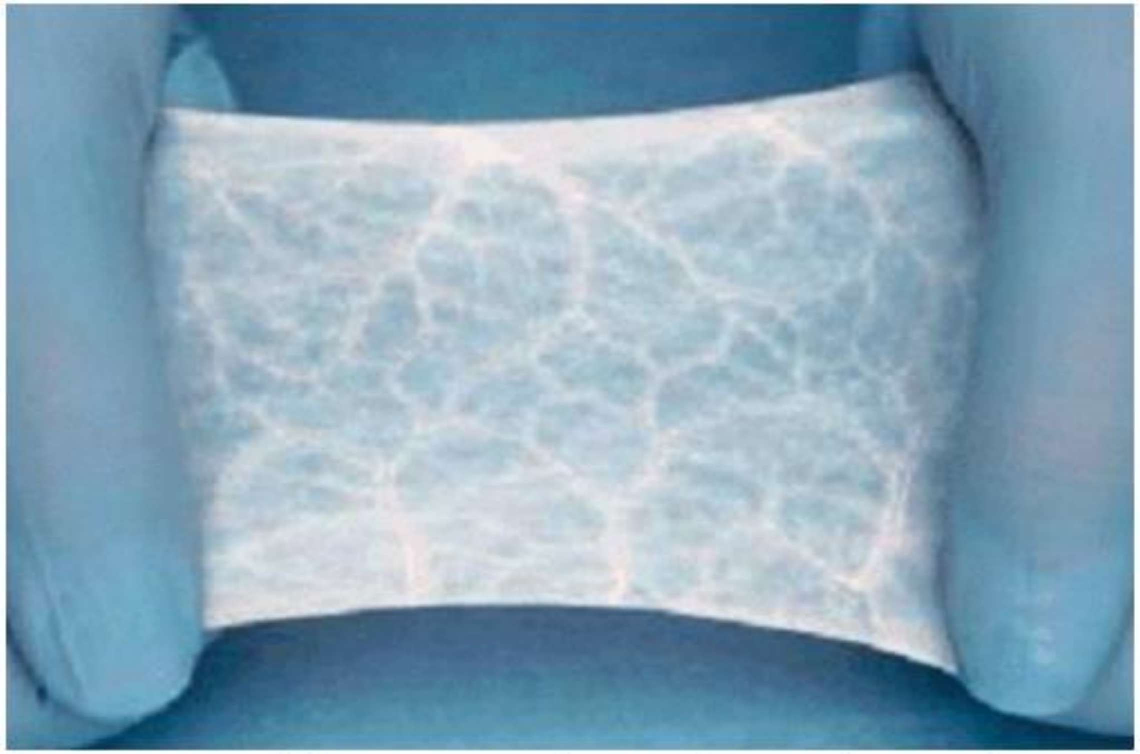


Fig. 145.11 An acellular xenogeneic dressing. This particular wound matrix is derived from porcine small intestine submucosa (Oasis®) and serves as a

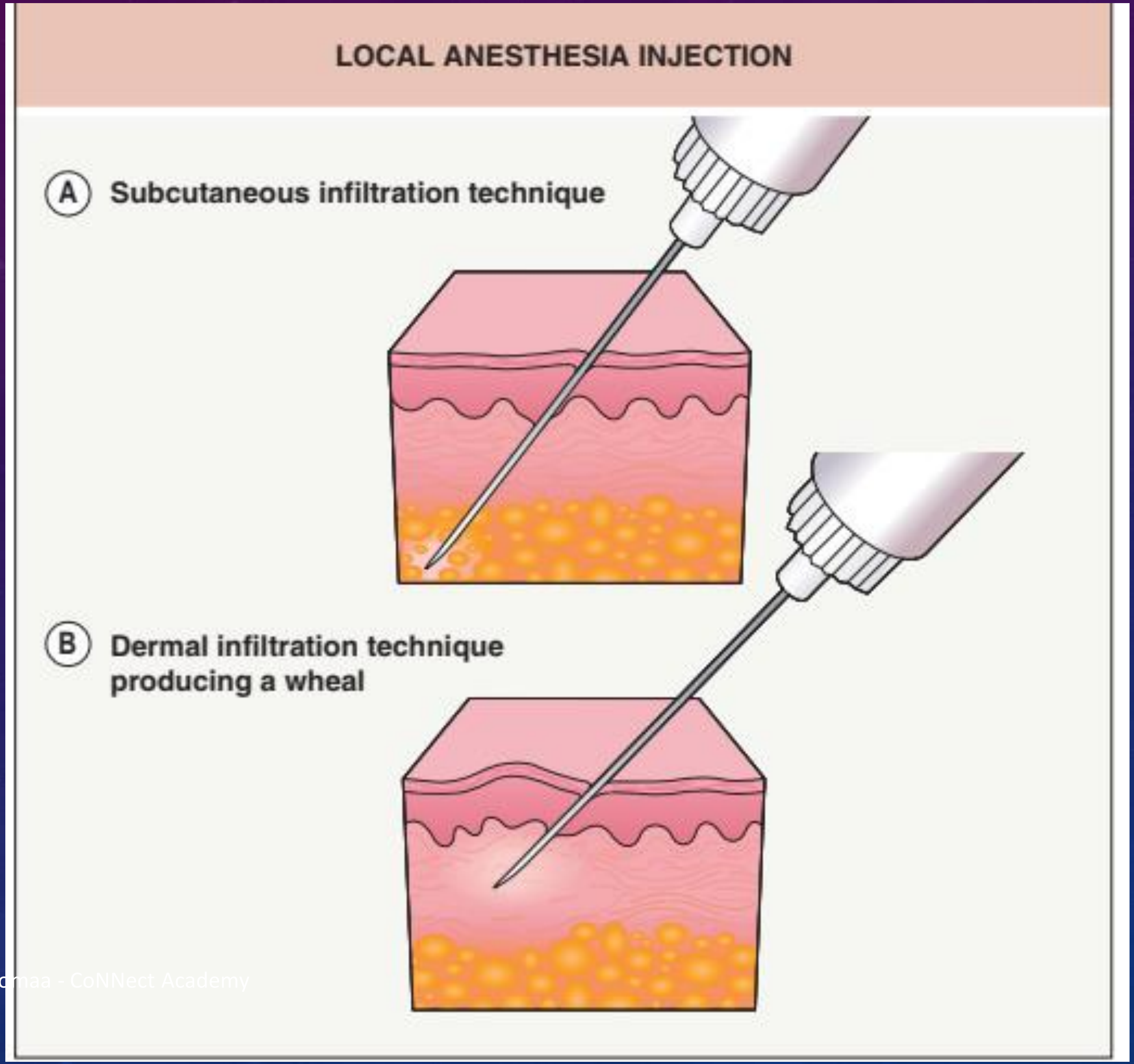
TYPES OF COMPRESSION SYSTEMS			
Compression system	**Examples**	**Advantages**	**Disadvantages**
Long stretch (elastic) bandages	ACE™ Biflex® Dauerbinde® K SurePress®	• Base of the toes to knees • Low working pressure • High resting pressure • 100–200% extensibility • Can be applied as spiral or figure-of-eight • Inexpensive; washable and reusable forms available	• Tend to unravel • Do not provide a sustained compression • Risk of incorrect application • Can lose elasticity with continued use
Short stretch (inelastic) bandages	Zinc paste (Viscopaste®/Unna boot) Action, Comprilan®, Panelast®, Porelast®	• Comfortable so better tolerated • Minimal interference with daily activities • Some (e.g. Viscopaste®) can generate high working pressure	• Can lose compression after application • Not good for highly exudative wounds • Viscopaste®/Unna boot needs to be applied by well-trained staff
Devices	FarrowWrap® (with Velcro®) CircAid® (with Velcro®)	• Adjustable compression • Easy to put on and remove	• Expensive • Bulky
Multicomponent bandages	Coban™ 2, Coban™ 2 Lite Profore® (5 layers), Profore® Lite (4 layers) Actico®2C	• Higher compression and sustained high compression • Graduated compression • Reduced ("Lite") compression available for patients with mixed venous and arterial disease	• Needs to be applied by well-trained staff
Intermittent compression devices	Lympha Press® (pneumatic pump)	• Enhance fibrinolytic activity	• Expensive • Requires being immobile for a few hours per day
Tubular system	Tubigrip™ EdemaWear®	• Easy to use • Tubular bandage doubled back over limb to increase compression	• Low compression (unless that is what is desired because of coexisting arterial disease)

Fig. 145.14 Unna boot. Nonelastic compression therapy with a zinc-

BIOPSY TECHNIQUES AND BASIC EXCISIONS

SELECTION OF TYPE OF BIOPSY TECHNIQUE				
Method	**Indication**	**Type of specimen obtained**	**Anesthetic technique (see Fig. 146.1)**	**Closure**
Curettage	• Lesions involving the epidermis (e.g. SKs, AKs, verrucae) • Confirm clinical diagnosis of BCC prior to definitive treatment	Epidermal sheet or fragments of epidermis and dermis	Wheal	Secondary
Scissors biopsy	• Pedunculated lesion	Tissue above connection to the epidermis	None or wheal	Secondary
Shave biopsy	• Lesions involving the epidermis ± superficial dermis • Elevated lesion	Epidermis, papillary dermis and, occasionally, reticular dermis (elevated lesions)	Wheal	Secondary
Saucerization biopsy	• Lesions involving the epidermis and dermis • Atypical melanocytic nevus versus thin melanoma • Hypertrophic AK versus minimally invasive SCC	Similar to a shave biopsy, but specimen is thicker (i.e. contains reticular dermis)	Wheal or deep infiltration	Secondary
Punch biopsy	• Process (e.g. tumor, inflammation) involves the dermis • Depressed lesion	Epidermis, dermis and, sometimes, subcutaneous fat	Wheal or deep infiltration	Primary; simple suture
Incisional biopsy	• Process (e.g. tumor, inflammation) involves the subcutaneous fat or fascia • Large-sized tumors • Subtle disorders of dermal connective tissue	Epidermis, dermis and subcutaneous fat Can include fascia where needed	Deep infiltration	Primary; layered closure
Excision *in toto*	• Biopsy intended to be definitive treatment • Strong clinical suspicion of cutaneous malignancy (e.g. invasive melanoma)	Epidermis, dermis and subcutaneous fat Can include fascia where needed	Deep infiltration	Primary; layered closure

SPECIMEN HANDLING		
Proposed laboratory test	**Carrier medium**	**Comments**
Routine microscopy, immunohistochemistry, PCR assay	10% neutral buffered formalin	Fixation process begins immediately
Direct immunofluorescence	Michel's medium or fresh*	Depends on laboratory preference or availability
Flow cytometry	Fresh*	Lymphoma cutis
Culture for bacteria, mycobacteria or fungi	Fresh* or minced in sterile culture/ carrier medium appropriate for organism (usually performed by laboratory)	
Culture for viruses	Viral transport medium (e.g. M4RT®)	
Electron microscopy	Glutaraldehyde	


LOCAL ANESTHESIA INJECTION
A Subcutaneous infiltration technique
B Dermal infiltration technique
producing a wheal

ANTISEPTIC AGENTS		
Agent	**Advantages**	**Disadvantages**
Povidone-iodine (Betadine®)	Broad antimicrobial spectrum, including fungi	Irritant and allergic contact dermatitis Residual color May cross-react with iodine in radiocontrast media and iodides in medications
Chlorhexidine (Hibiclens®)	Broad antimicrobial coverage Limited systemic absorption Prolonged suppression of bacterial growth	Keratitis due to ocular exposure Cochlear damage if enters middle ear Irritant and allergic contact dermatitis Contact urticaria Very rarely, anaphylaxis
Isopropyl alcohol	Inexpensive Denatures protein, including bacterial cell walls Immediate (but not prolonged) effect	Weak antimicrobial activity Flammable in the setting of cautery Skin irritant Skin must remain wet for 2 minutes for maximum effect
Chlorhexidine–isopropyl alcohol combination (ChloraPrep®)	Combination provides both short-term and long-term effects Evidence it is more effective than povidone-iodine[10]	Those of either agent alone (see above)
Hexachlorophene (pHisoHex®)	Strong effect against Gram-positive cocci	Little effect on Gram-negative organisms or fungi Teratogen Absorbed through skin with potential neurotoxicity in infants
Soap and water	Traditional preoperative handwashing regimen Disinfects hands as well as alcohol-based rubs[62] Better against *Clostridium difficile* and Norwalk virus	At operative site, cumbersome and messy No prolonged antisepsis
Hydrogen peroxide	Readily available Inexpensive	No significant antiseptic properties Cytotoxic to keratinocytes *in vitro*

TECHNIQUES

TECHNIQUES

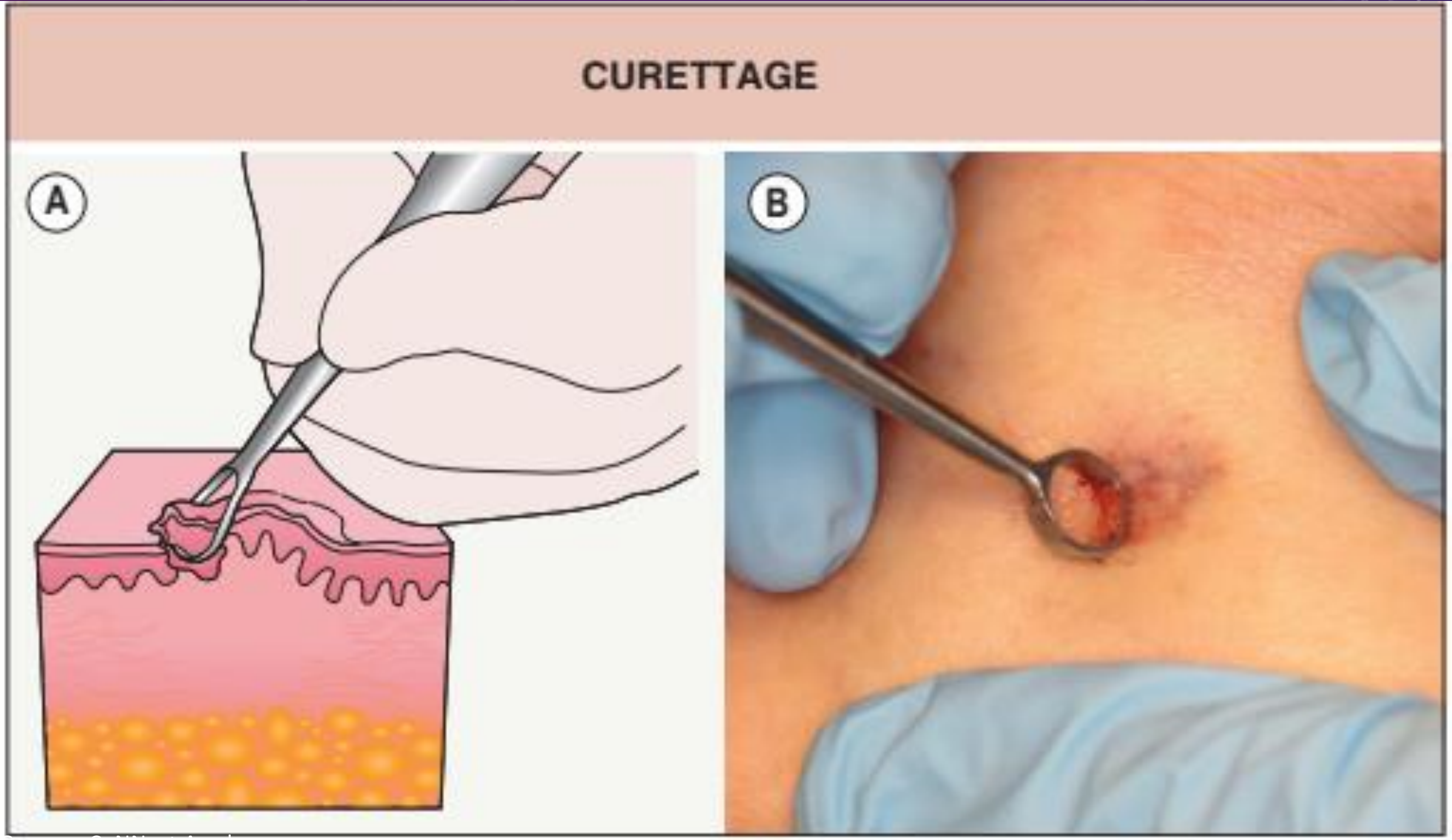


Fig. 146.5 Curettage. **A** Diagram. **B** Demonstration of technique.

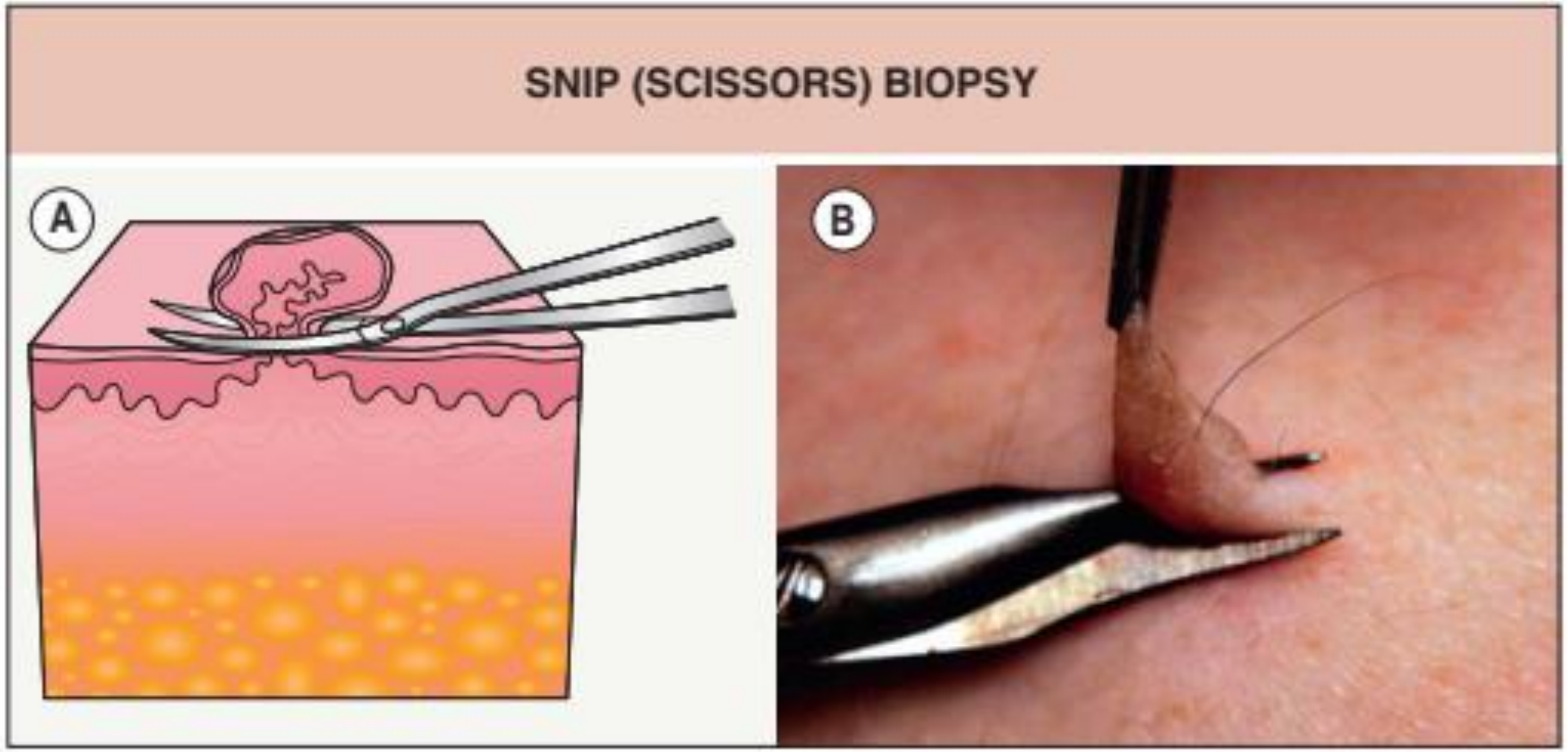


Fig. 146.6 Snip (scissors) biopsy. A Diagram. B Demonstration of technique.

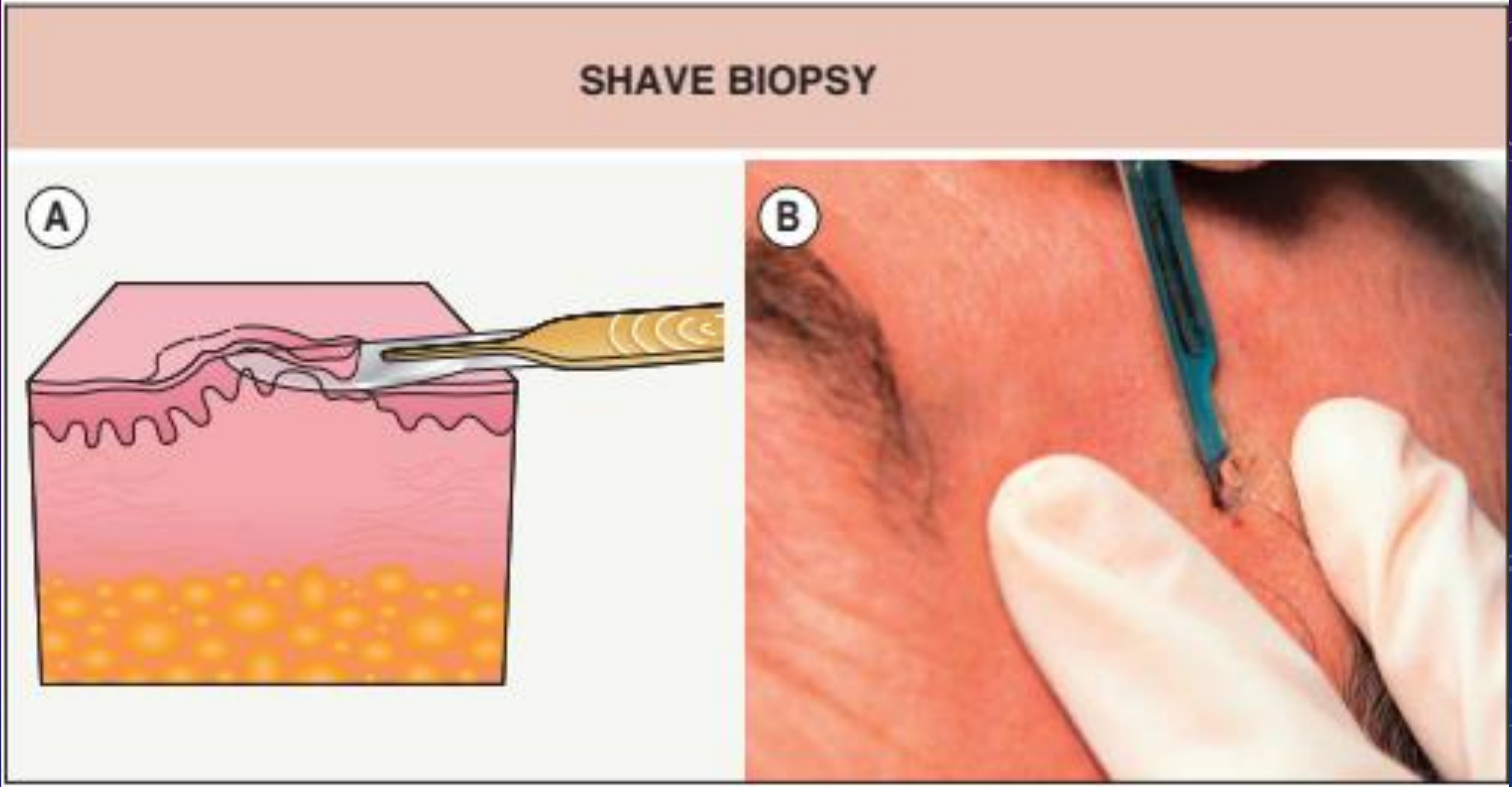


Fig. 146.7 Shave biopsy. A Diagram. B Demonstration of technique.

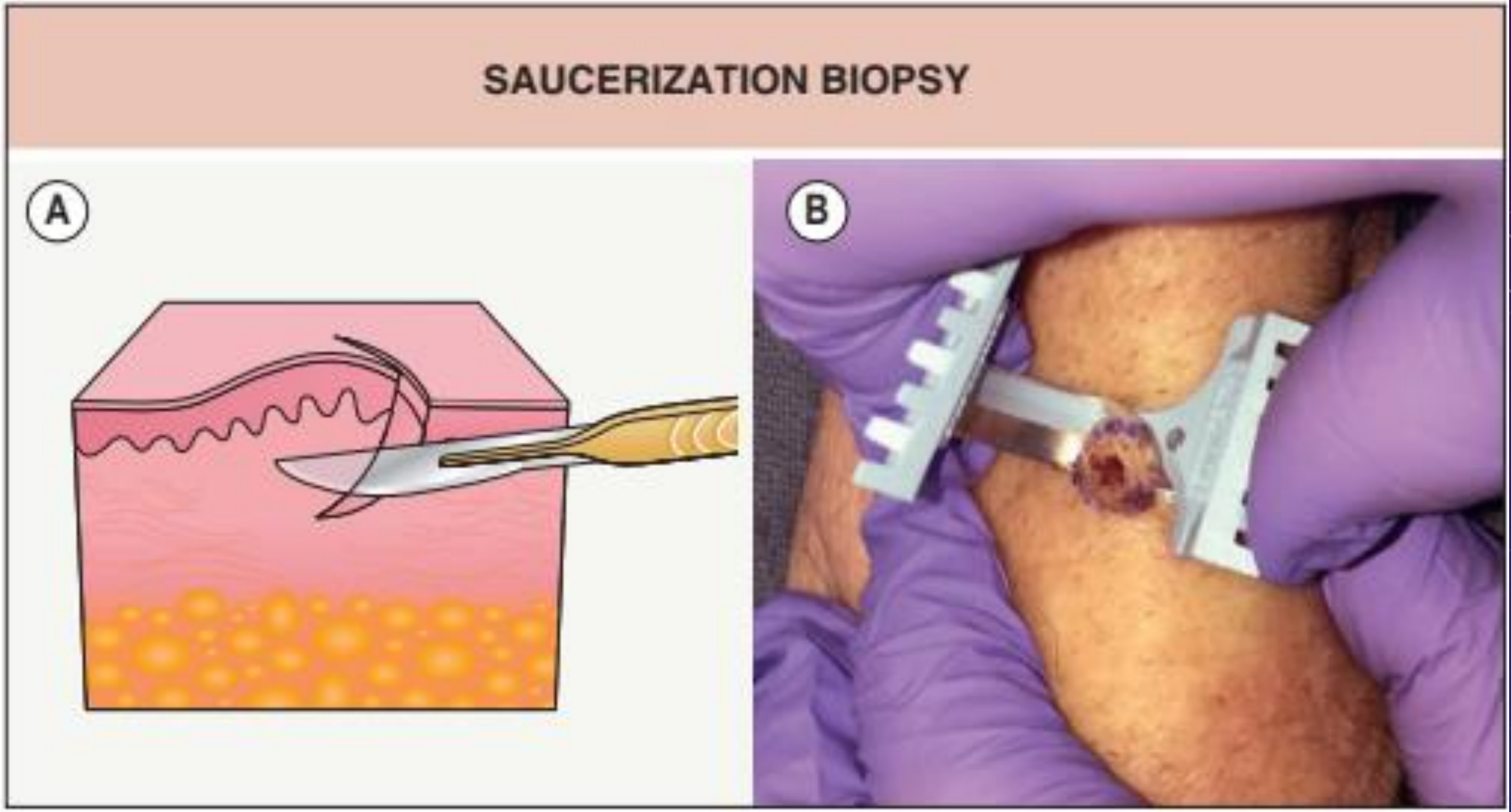


Fig. 146.8 **Saucerization biopsy. A** Diagram. **B** Demonstration of technique.

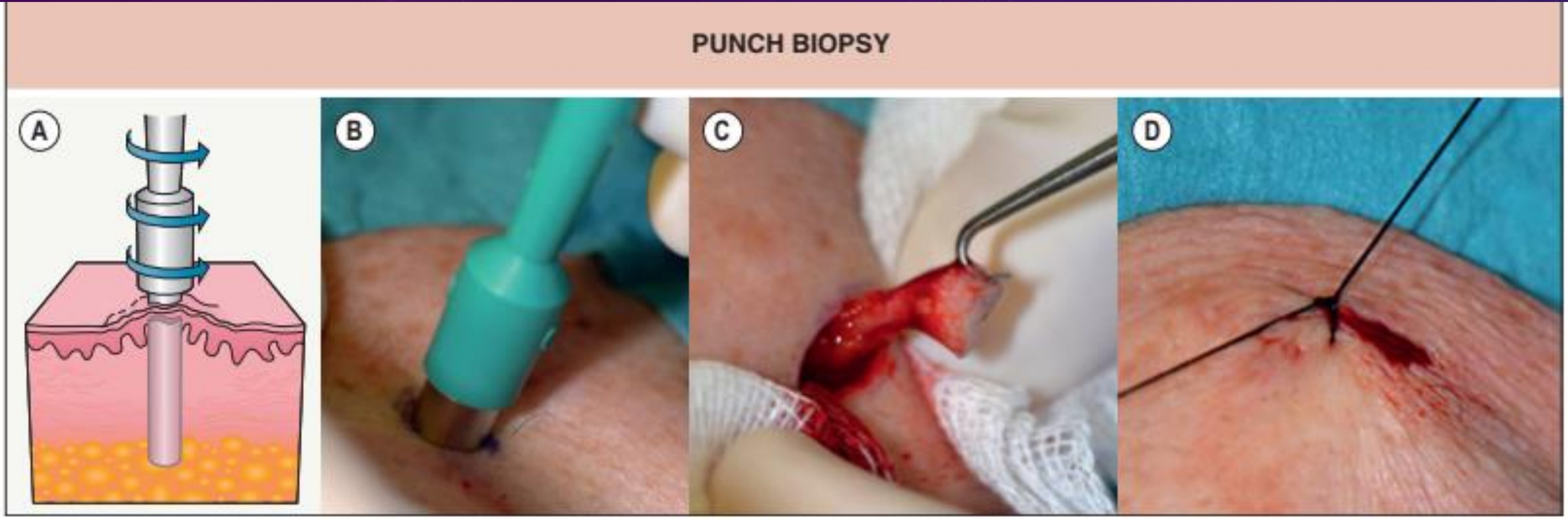


Fig. 146.9 Punch biopsy.
A Diagram. **B** Punch biopsy instrument rotated into the skin. **C** The biopsy specimen is gently removed with a skin hook placed peripherally and superficially in order to avoid crush artifact. **D** Closure of the biopsy wound with a simple epidermal stitch.
Photographs, Courtesy, Luis Requena, MD.

SPECIAL CONSIDERATIONS FOR A PUNCH BIOPSY ON THE FACE

- Position patient in a supine position with draping or gauze to protect the eyes from light or inadvertent damage by suture or instrument
- Use a 3 mm punch biopsy instrument (if possible); a 4 mm punch biopsy wound can produce standing cones when closed
- Choose a site with low visibility (if possible), e.g. in the shadows of the face, at cosmetic unit boundaries, or off/beyond the center of the face
- Protect underlying tissue with an anesthetic wheal
- Tailor depth of excision to depth of pathology rather than rotating and pushing the punch biopsy instrument to its hub
- Control bleeding with suture(s) – simple interrupted stitches of monofilament 5–0 or 6–0 nylon or polypropylene
- Protect with bandage and remove sutures at 5–7 days

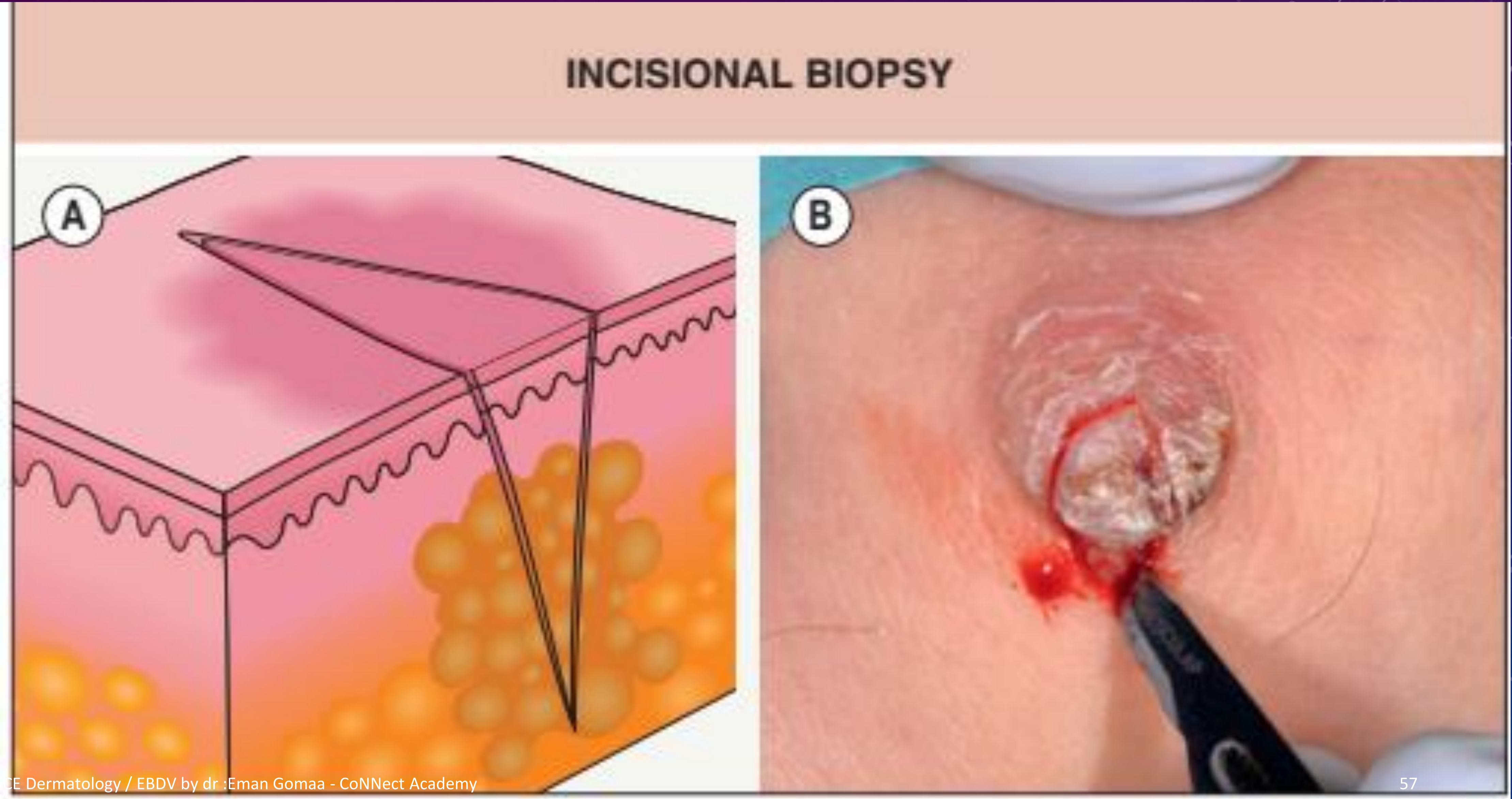


Fig. 146.10 **Incisional biopsy. A** Diagram. **B** Demonstration of the procedure.

Fig. 146.12 Fusiform excision of a squamous cell carcinoma. A,B After the lesion and appropriate margins are marked, the site is anesthetized, prepped with an antiseptic agent, and draped with sterile towels. **C** Stabilizing the site with traction, the epidermis on one side of the fusiform design is scored using a #15 blade and then the opposite side is scored. **D** The incision is completed into the appropriate plane in the subcutaneous tissue and the specimen then sits up in the middle of the wound like an island. **E,F** The base of the specimen is dissected with scissors or a blade. **G,H** The wound edges are then undermined in the same plane as the base of the wound with blunt-tipped scissors or a blade. Electrodesiccation or electrocoagulation is used to address small actively bleeding vessels to achieve hemostasis. **I,J** The subepidermal space is closed with buried sutures. **K,L** The epidermal edges are apposed by simple interrupted sutures.

SHORT-TERM AND LONG-TERM COMPLICATIONS OF BIOPSY PROCEDURES AND EXCISIONS

Short-term complications

- Irritant or allergic contact dermatitis to topical antibiotics and/or tapes and bandages
- Hemorrhage
- Poor wound healing; dehiscence of closed wounds
- Pain
- Infection
- Pruritus, paresthesias

Long-term complications

- Atrophic scars (thin, depressed, spread)
- Hypertrophic scars (red, thickened)
- Keloids
- Distortion of normal structures (e.g. ectropion, elevated nasal ala, notching of the lip or helix of the ear)
- Hypoesthesia, anesthesia
- Paralysis (e.g. due to injury to facial nerve branches)